

GRNsight v7: a web application for visualizing gene regulatory networks and protein-protein interactions

Ngoc Kim Ngan Tran & Milka Zekarias
Department of Computer Science
Loyola Marymount University

Southern California Conference
for Undergraduate Research
November 22, 2025



Loyola Marymount University
Frank R. Seaver
College of Science
and Engineering

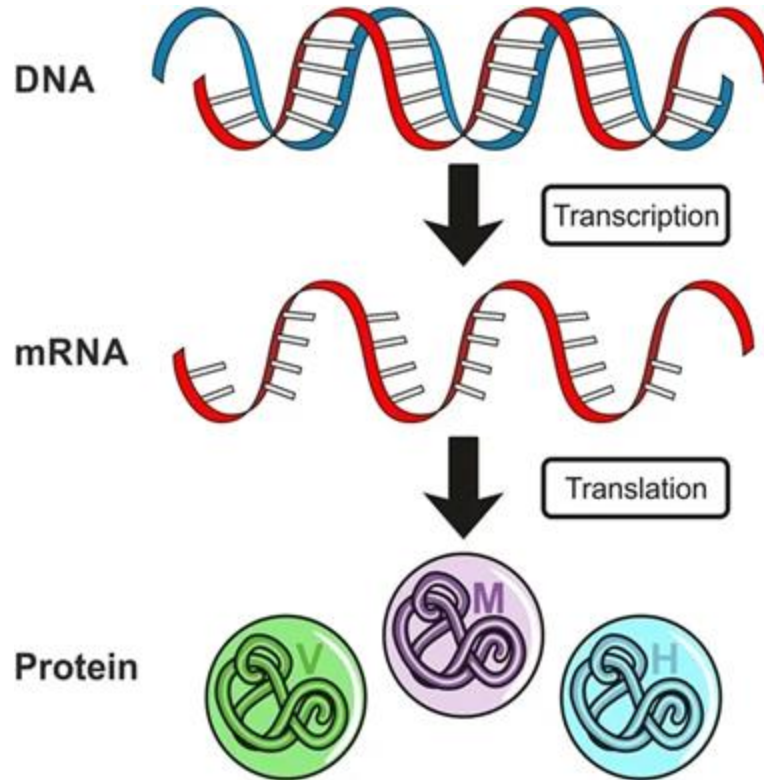
Outline

- Gene Regulatory Networks (GRN) consist of genes, transcription factors, and the regulatory connections between them.
- Protein-Protein Interactions (PPI) are the binding relationships between proteins.
- GRNsight is an open source tool that allows for easier visualization of GRNs and PPIs.
- We transitioned our database source from YeastMine to AllianceMine.
- GRNsight has improved PPI support for node coloring.
- GRNs have directed edges to indicate regulatory relationships (activation or repression).
- PPIs have undirected edges to indicate physical binding.
- New graph functionality allows users to restrict the graph to the viewport.
- GRNsight switched CI tools from Travis CI to Github Actions.

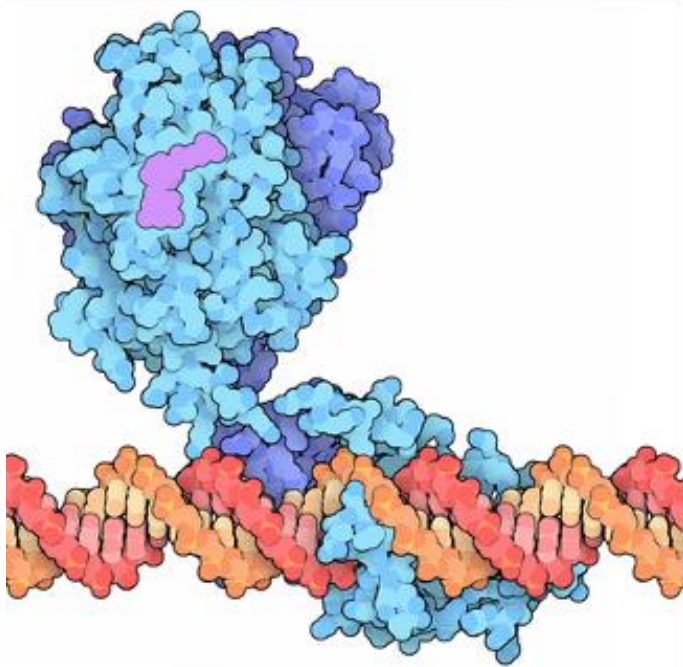
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Gene expression is the flow of information in a cell from DNA to RNA to protein



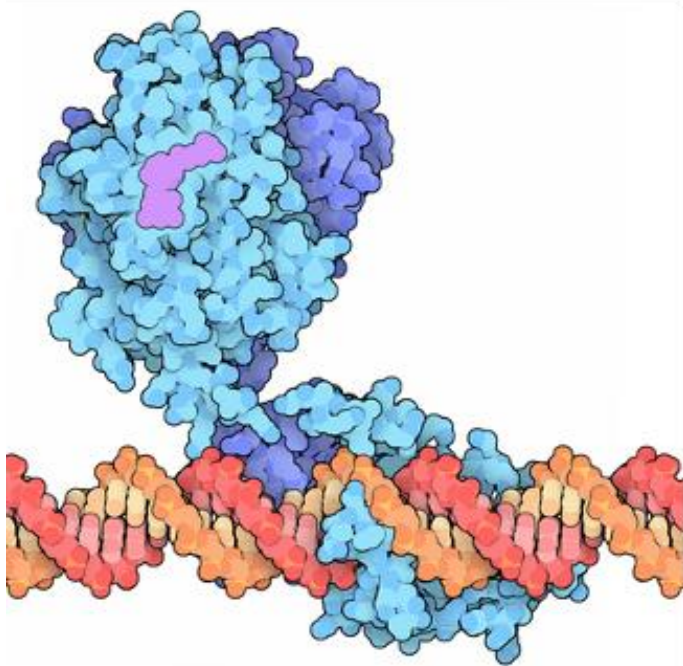
Regulatory Transcription Factors bind to DNA and turn on (activate) or turn off (repress) gene expression



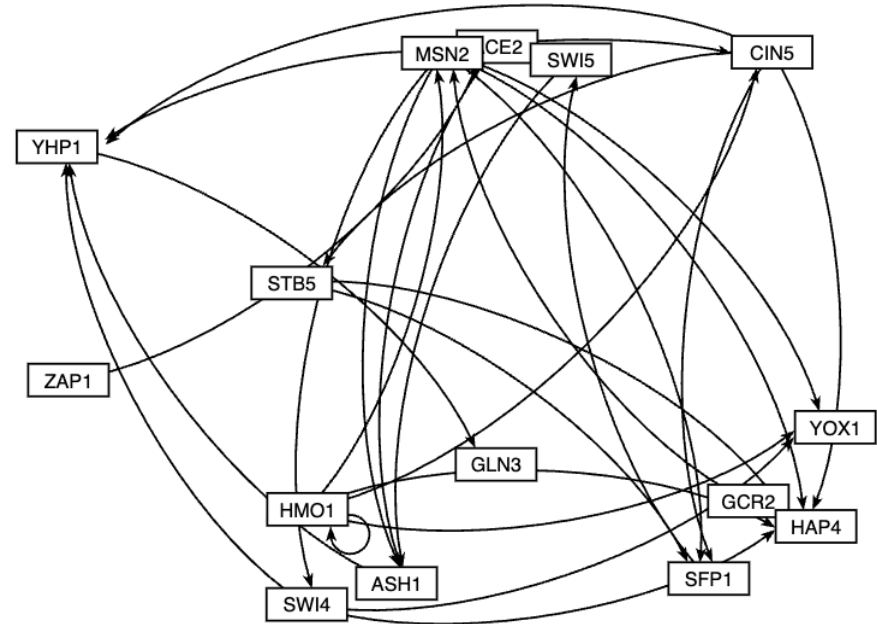
- In yeast, ~250 regulatory transcription factors regulate ~6000 genes.
- In humans, ~500 regulatory transcription factors regulate ~20,000 genes.

"PDB101: Molecule of the Month: Vitamin D receptor," RCSB: PDB-101.
<https://pdb101.rcsb.org/motm/155>

A Gene Regulatory Network (GRN) consists of genes, transcription factors, and the regulatory connections between them which govern the level of expression of mRNA and protein from genes



"PDB101: Molecule of the Month: Vitamin D receptor," RCSB: PDB-101.
<https://pdb101.rcsb.org/motm/155>



Protein-Protein Interactions refer to the physical interactions of proteins binding to each other - the basis of nearly all cell functions

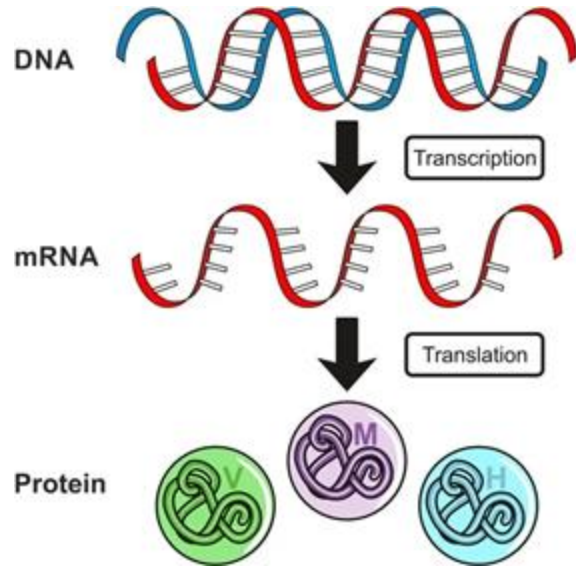
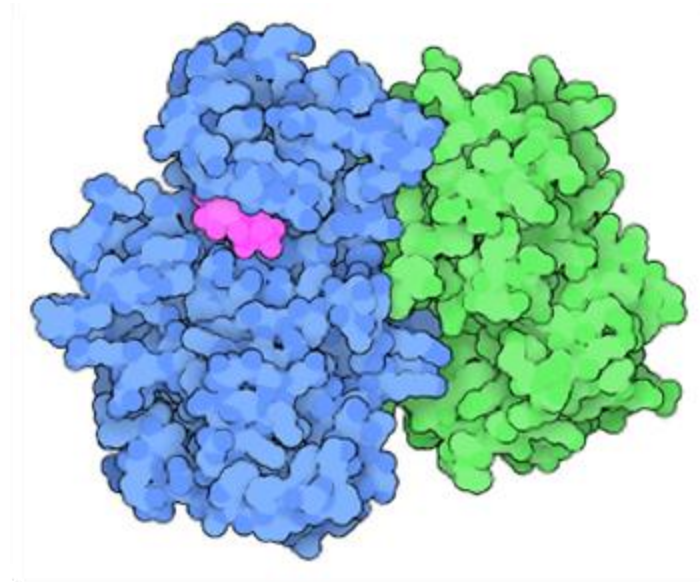


Image from Dr. Surat P, Ph.D. on AZO Life Sciences



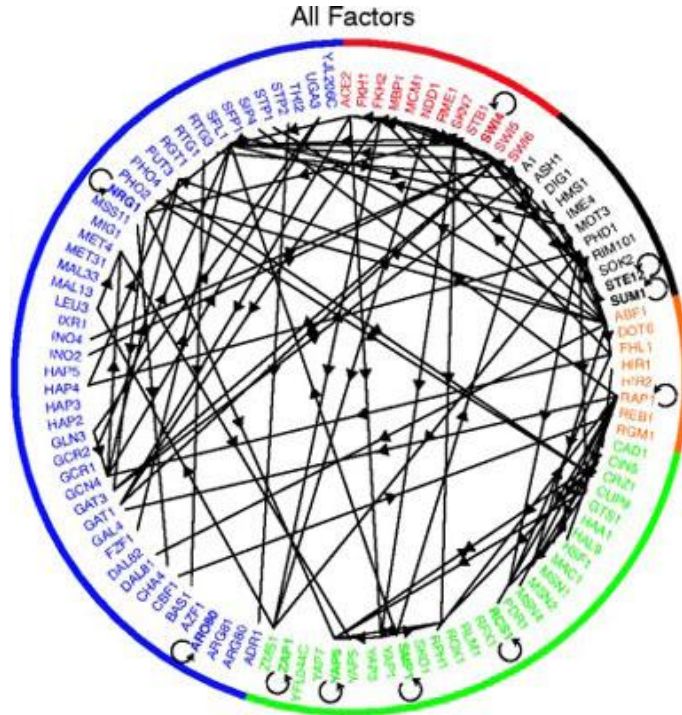
"PDB101: Molecule of the Month: Cyclin and cyclin-dependent kinase,"
RCSB: PDB-101. <https://pdb101.rcsb.org/motm/236>

Outline

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Large gene regulatory networks are difficult to visualize

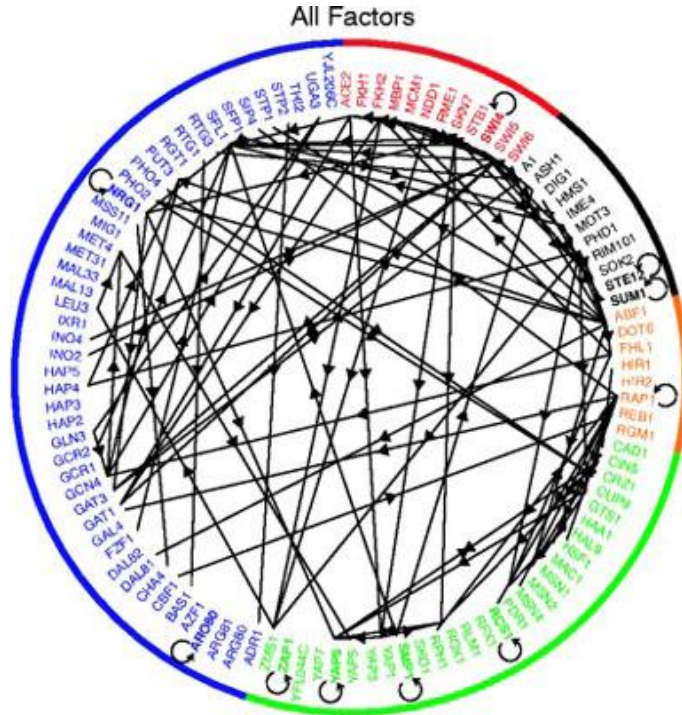
Gene Regulatory Network (GRN) of 106 Transcription Factors in Yeast



Lee et al. (2002)

Large gene regulatory networks are difficult to visualize

Gene Regulatory Network (GRN)
of 106 Transcription Factors in Yeast



Lee *et al.* (2002)



GRNsight rapidly generates graphs using customizations to the Open Source D3 Library



GRNsight

Web app and service for visualizing models of gene regulatory networks.

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GRNsight v7.2.0

Welcome to GRNsight. Please select a file to upload.

Network Layout Node Edge View Export Help Demo Demo #1: Unweighted GRN (15 genes, 28 edges)

15 nodes
28 edges

Network

Network Source

Demo

Demo #1: Unweighted



Open File



Load from Database

Reload

Network Mode

Gene Regulatory Netw

Species:

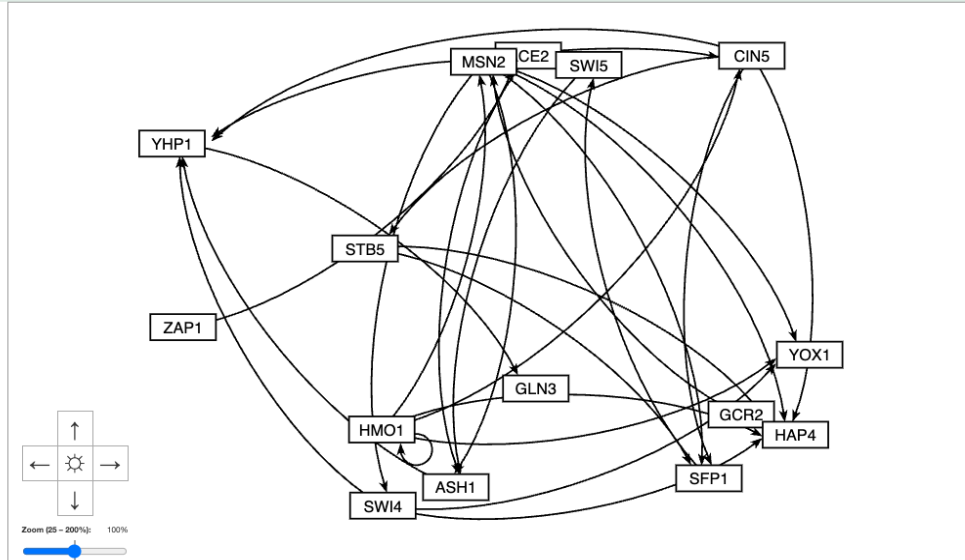
Saccharomyces cerevisiae

Layout

Node

Edge

View



To get started quickly, GRNsight also provides different demos



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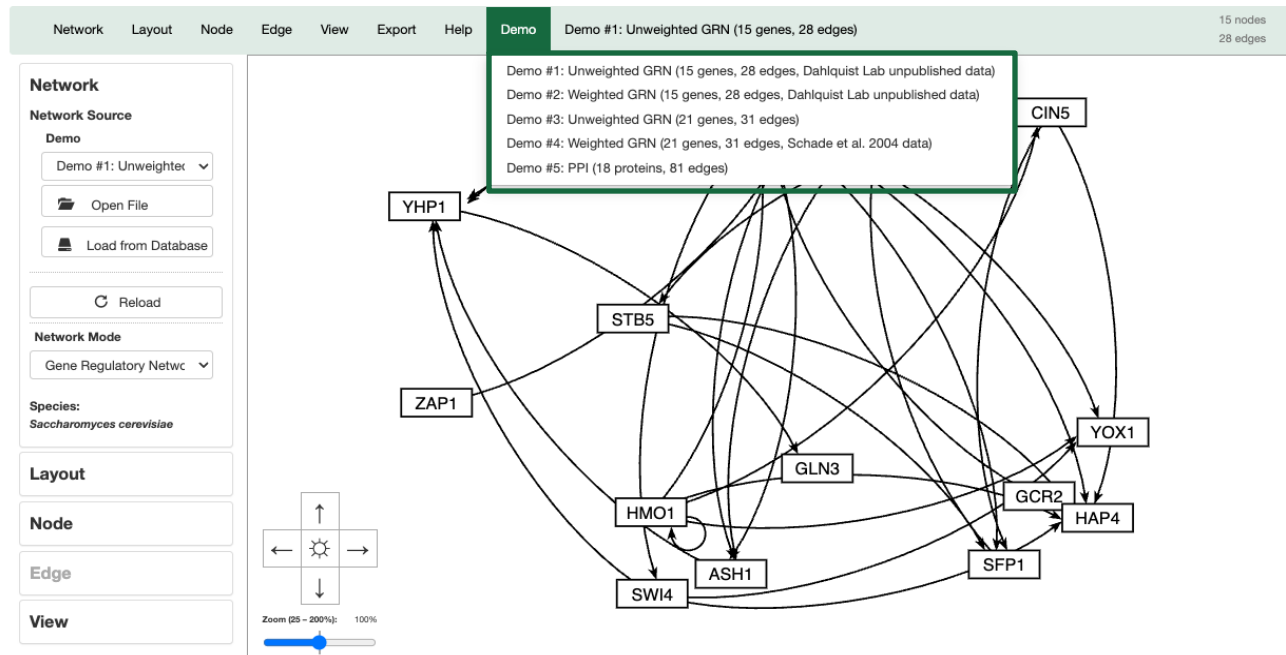
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Users can import networks via the file upload functionality



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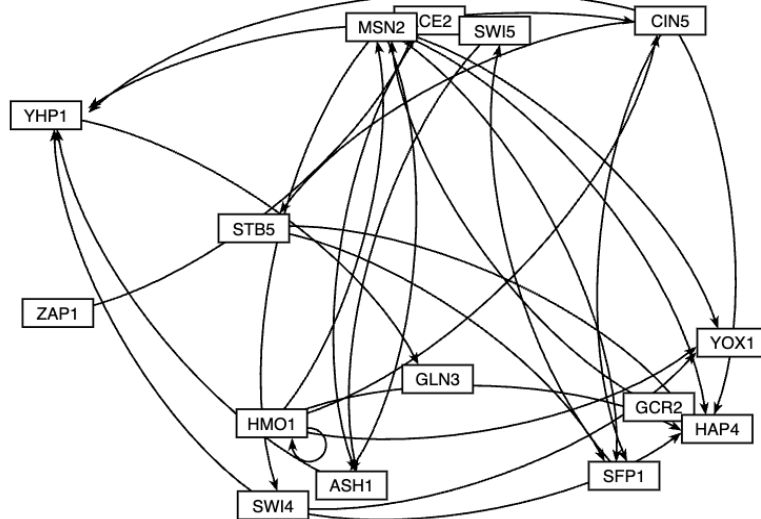
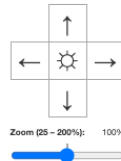
Saccharomyces cerevisiae

Layout

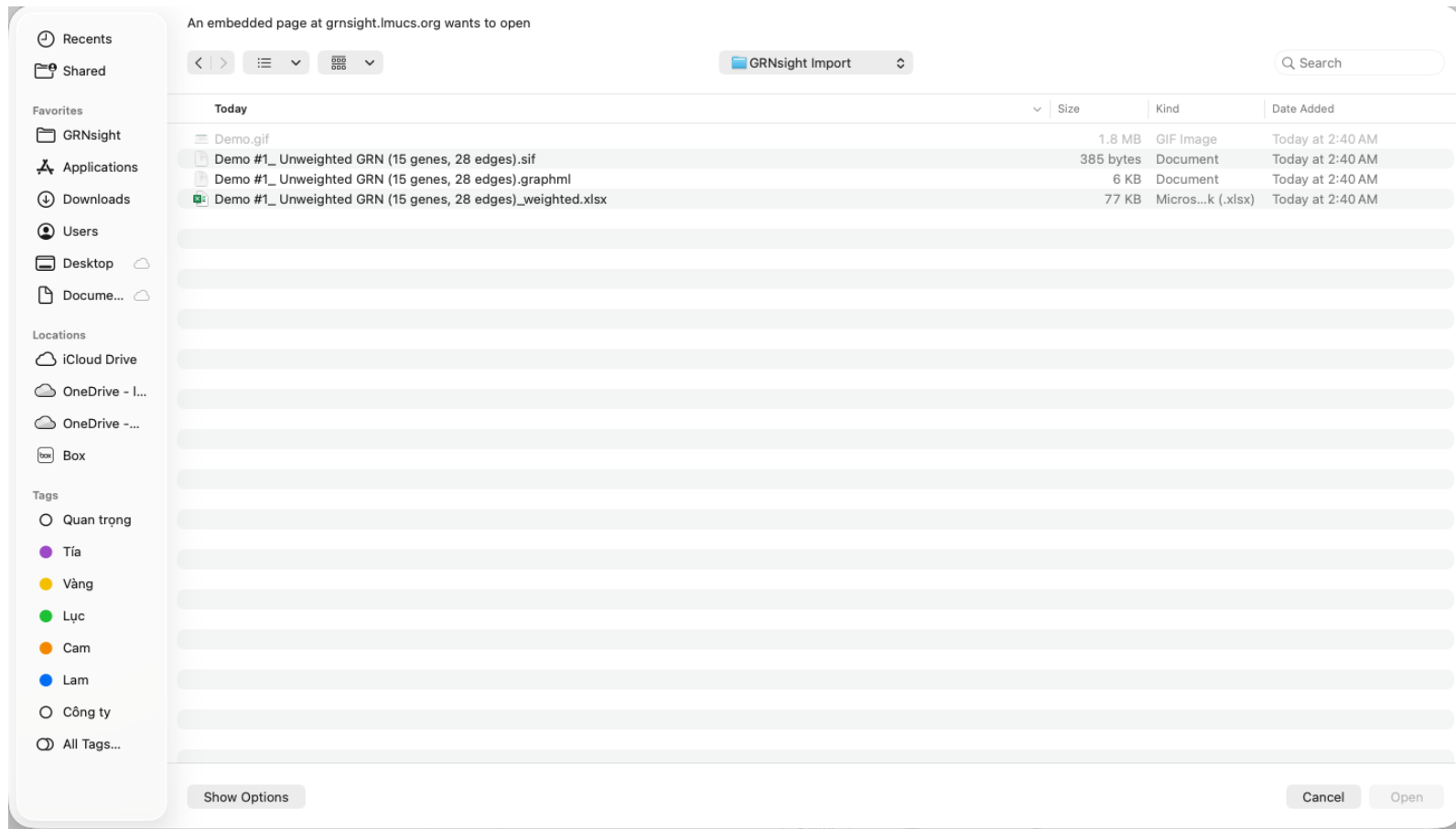
Node

Edge

View



The imported file needs to be in a supported format (SIF, GraphML, or Excel)



Users can load networks from our database, which contains regulation data from the *Saccharomyces* Genome Database (SGD)



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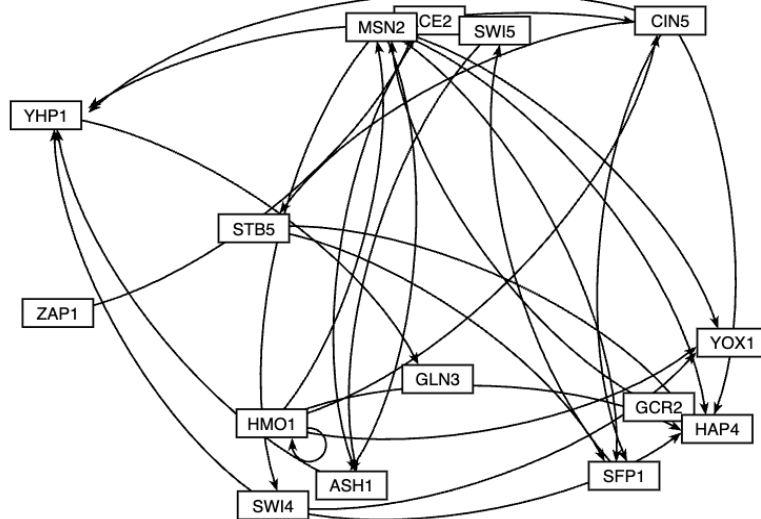
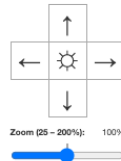
Saccharomyces cerevisiae

Layout

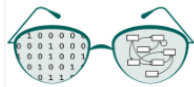
Node

Edge

View



Users can load different network types, either GRN or PPI



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Load from Database

Reload

Network Mode

Gene Regulatory Network

Species:

Saccharomyces cerevisiae

Layout

Node

☐ Enable Node Coloring

Edge

View

Zoom (25 - 200%) 100%

Load from Database

Generate Network

Warning: changing network type or source will clear the list of selected genes or proteins below.

Network Type ☒ Gene Regulatory
☐ Protein-Protein Physical Interactions

Network Source yeastmine - SGD: 2024-03-19

Select gene

Added genes go below! Click on a gene to remove it.

Users can choose different network sources since GRNsight refreshes with new data each year



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Demo

Demo #1: Unweighted

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Load from Database

Reload

Network Mode

Gene Regulatory Network

Species:

Saccharomyces cerevisiae

Layout

Node

Enable Node Coloring

Edge

View

Zoom (25 ~ 200%): 100%

Load from Database

Generate Network

Warning: changing network type or source will clear the list of selected genes or proteins below.

Network Type Gene Regulatory

Network Source Yeastmine - SGD: 2024-03-19
Yeastmine - SGD: 2023-03-08
Yeastmine - SGD: 2022-03-07

Select gene

Added genes go below! Click on a gene to remove it.

Generate Network Cancel

CIN5

YOX1

HAP4

GCR2

SFP1

GLN3

ASH1

HMO1

SWI4

Starting in 2025, AllianceMine is GRNsight's new database source because YeastMine is no longer supported

YeastMine shutting down July 15

July 01, 2024

Due to ongoing cuts to our funding, SGD can no longer continue to provide the YeastMine data warehouse resource. It is with heavy hearts that we discontinue this service.

Back in 2011, SGD implemented InterMine (<http://www.InterMine.org>), an open source data warehouse system with a sophisticated querying interface, to create YeastMine, a multifaceted search and retrieval environment that provided access to diverse data types. YeastMine served as a powerful search interface, a discovery tool, a curation aid, and a complex database presentation format.

YeastMine has served us all quite well. We are working to move the YeastMine data into AllianceMine, hosted by the Alliance of Genome Resources, of which SGD is a founding member.

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Nodes can be colored via an on/off toggle



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Node Coloring Off

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Network

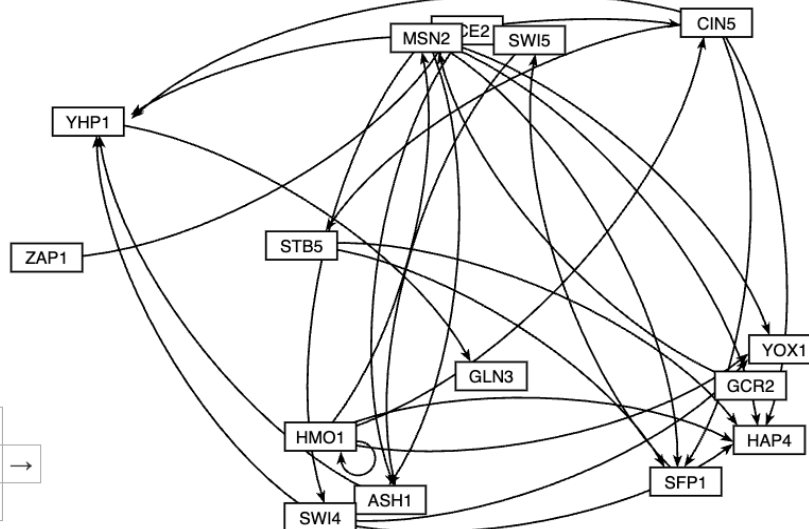
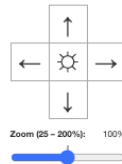
Layout

Node

☐ Enable Node Coloring

Edge

View



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GRNsight v7.2.0

Node Coloring On

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Network Layout Node Edge View Export Help Demo Demo #1: Unweighted GRN (15 genes, 28 edges)

Network

Layout

Node

☒ Enable Node Coloring

Select from user-uploaded
expression data, or use data
from our Expression Database

Top Dataset

wt_log2_expression

☒ Average Replicate Values

Bottom Dataset

Same as Top Dataset

☒ Average Replicate Values

Log Fold Change Max Value
(0.01-100):

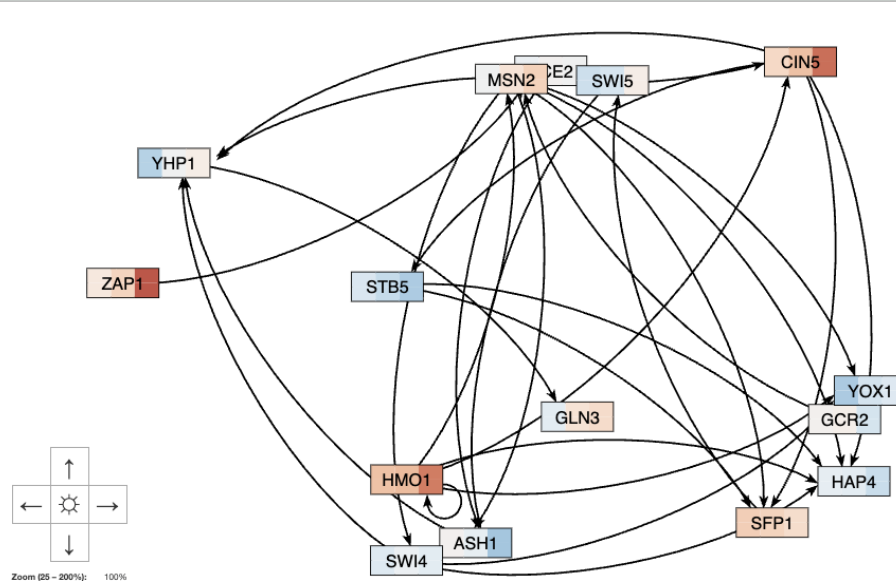
3

Set

-5.00 0 5.00

Edge

View



Nodes are colored with a heat map overlay to visualize time series expression data

ACE2



**Node Coloring With
Expression Dataset of
Dalquist 2018**

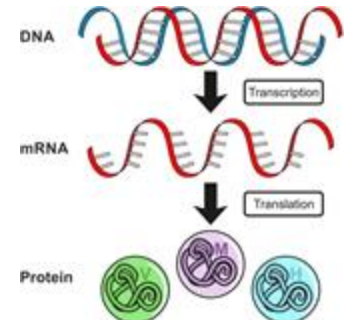
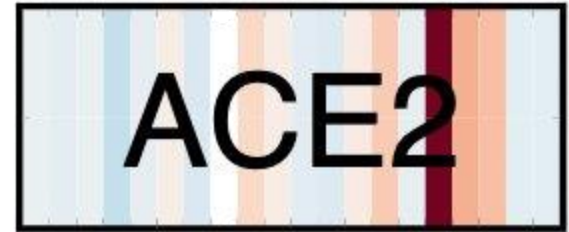


Image from Dr. Surat P, Ph.D. on
AZO Life Sciences

Each vertical slice on the node represents the replicate expression values of a transcription factor that were measured at each time point

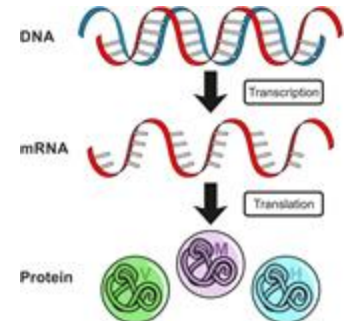


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Red slices represent positive expression values
(indicating an increase in mRNA)

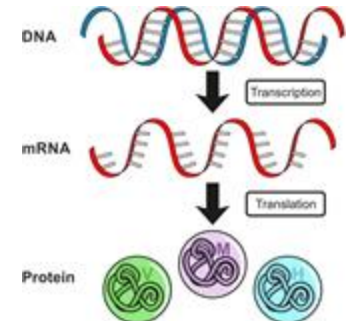


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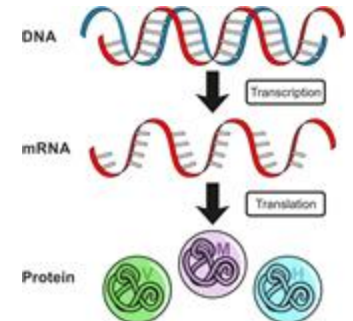
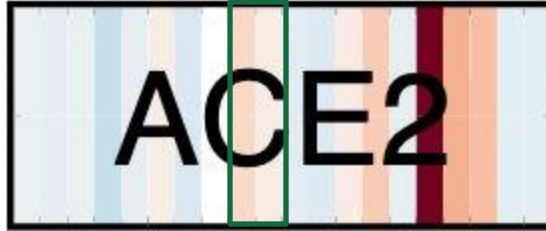


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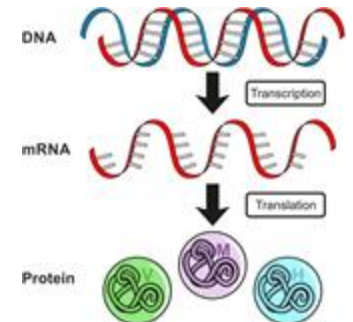
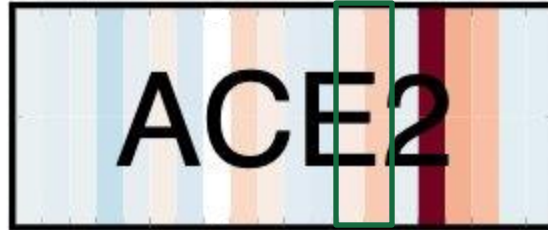


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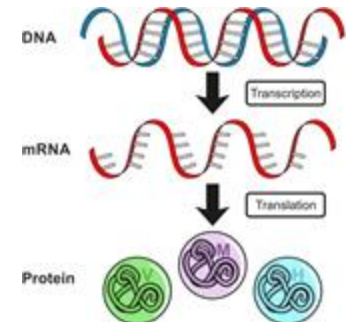
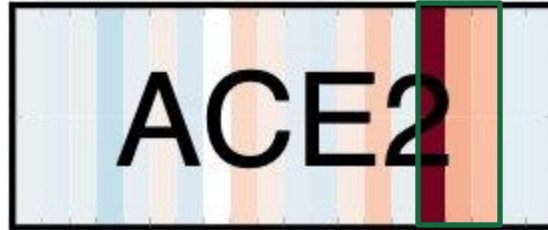
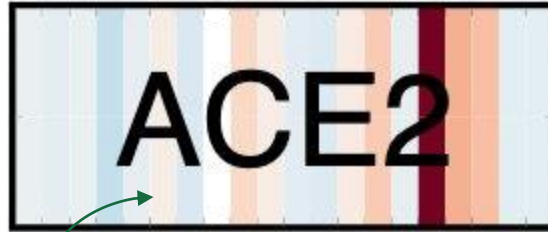


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The greater the magnitude of the expression value,
the darker the shade of red



Smaller magnitude of
increase in mRNA
expression at this
time point

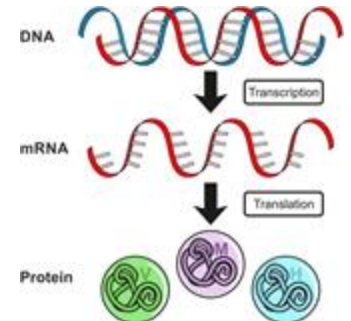
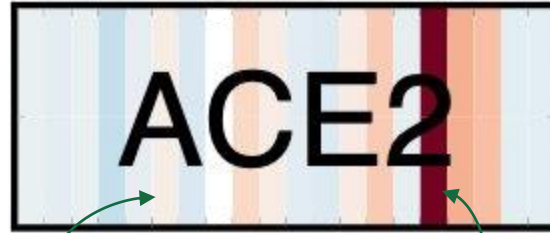


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The greater the magnitude of the expression value,
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Smaller magnitude of
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Greater magnitude of
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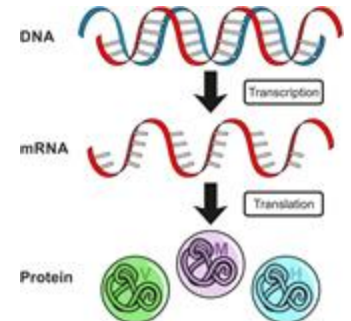


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Similarly, blue slices represent negative expression values (indicating a decrease in mRNA), with greater magnitudes as darker shades of blue

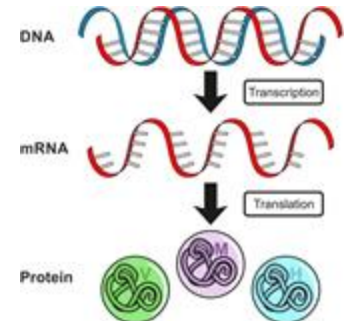
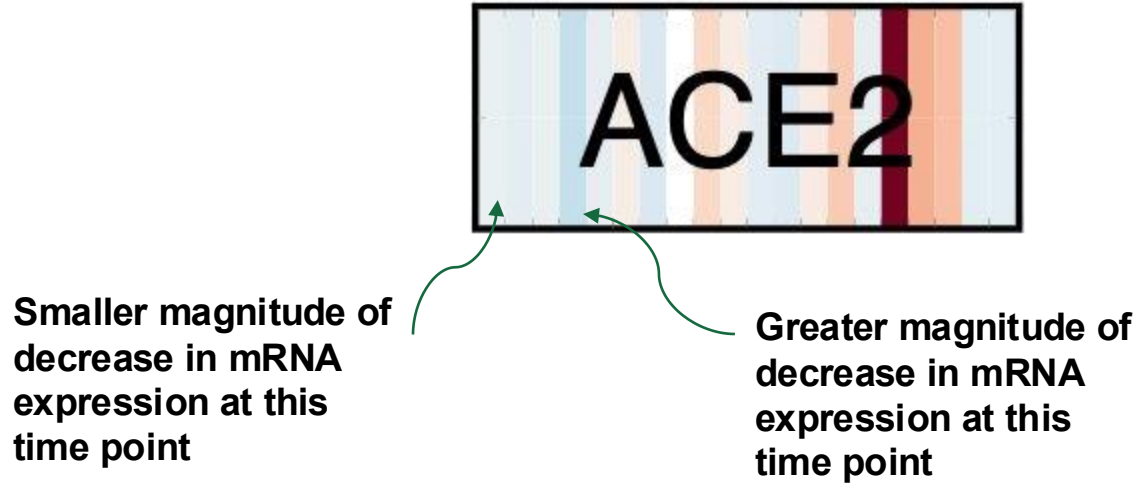


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Each node can display different datasets on the top and bottom

Node

☒ Enable Node Coloring

Select from user-uploaded expression data, or use data from our Expression Database

Top Dataset

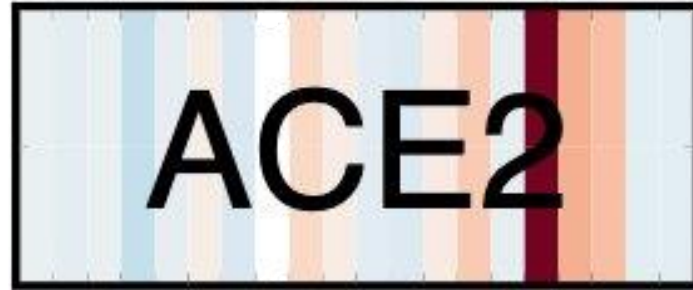
Dahlquist_2018_dcin5 ▾

☐ Average Replicate Values

Bottom Dataset

Same as Top Dataset ▾

☐ Average Replicate Values



Each node can display different datasets on the top and bottom

Node

☒ Enable Node Coloring

Select from user-uploaded
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from our Expression Database

Top Dataset

Dahlquist_2018_dcin5 ▼

☐ Average Replicate Values

Bottom Dataset

Thorsen_2007_wt ▼

☐ Average Replicate Values



When checking the "average replicate value" option, GRNsight will average replicate expression values taken at the same timepoint

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Select from user-uploaded
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from our Expression Database

Top Dataset

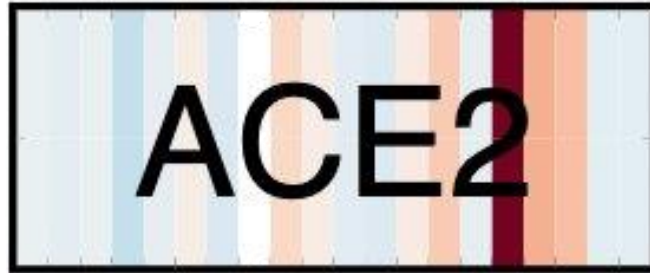
Dahlquist_2018_dcin5 ▼

☐ Average Replicate Values

Bottom Dataset

Same as Top Dataset ▼

☐ Average Replicate Values



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Dahlquist_2018_dcin5 ▼

☒ Average Replicate Values

Bottom Dataset

Same as Top Dataset ▼

☒ Average Replicate Values



Users can choose different expression datasets for node coloring



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Beta v7.3.3

Disclaimer: This is the beta version of GRNsight for live testing of new functionality. For the most stable version, please go to the GRNsight home page.

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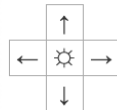
Demo

☒ wt_log2_expression

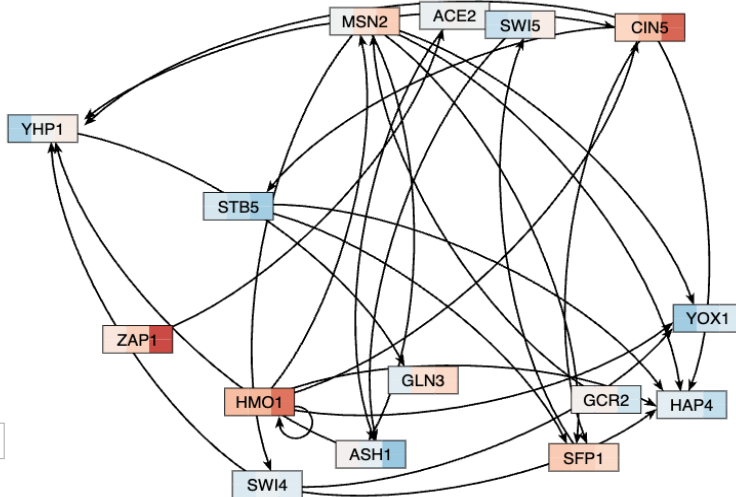
dcln5_log2_expression
dglN3_log2_expression
dhap4_log2_expression
dhmo1_log2_expression
dzap1_log2_expression

Expression Database

Apweiler_2012_wt
Barreto_2012_wt
Dahlquist_2018_dcln5
Dahlquist_2018_dglN3
Dahlquist_2018_dhap4
Dahlquist_2018_dzap1
Dahlquist_2018_wt
Kitagawa_2002_wt
Thorsen_2007_wt



Zoom (25 - 200%): 100%



Previously, node coloring was only enabled for GRNs because node coloring is a visualization of the level of mRNA expression



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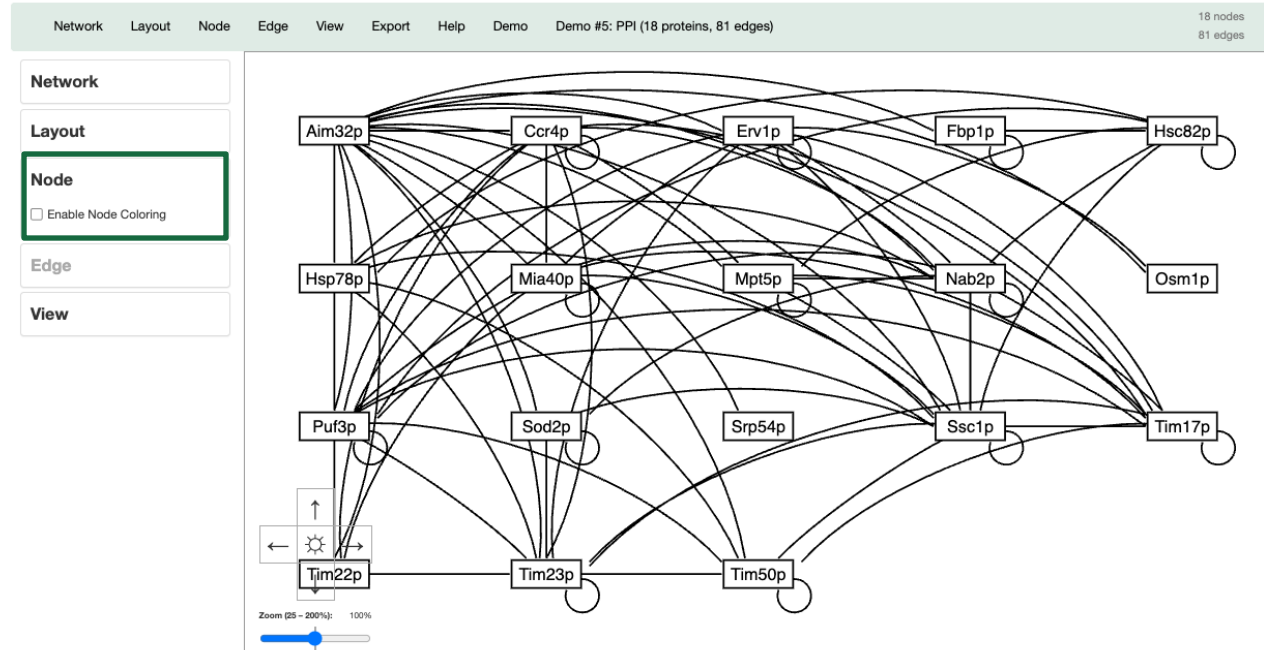
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12/4/2024

GRNsight v7.2.0 Node coloring is default to be off for PPIs

Welcome to GRNsight. Please select a file to upload.



Version 7.2 allows node coloring for PPIs but with warnings, as it is showing mRNA level data on a node representing a protein



GRNsight

Web app and service for visualizing models of gene regulatory networks.

A Joint Project of the
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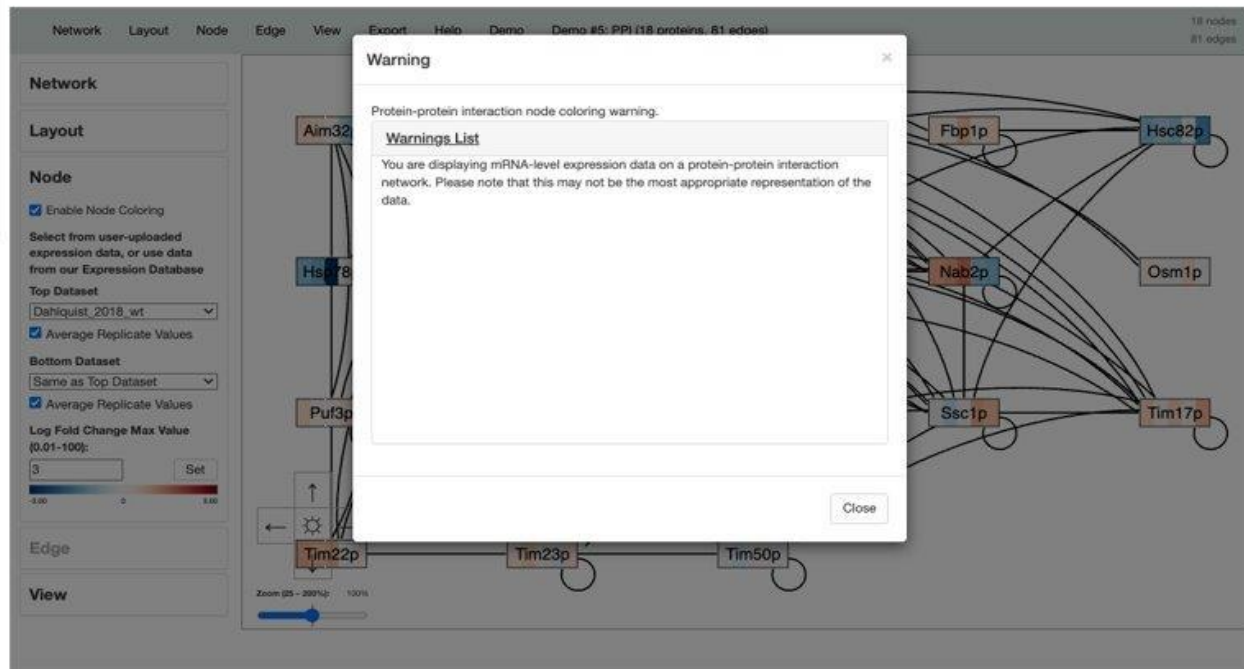
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GRNsight v7.2.0

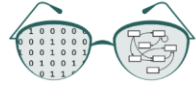
Welcome to GRNsight. Please select a file to upload.



Outline

- Gene Regulatory Networks (GRN) consist of genes, transcription factors, and the regulatory connections between them.
- Protein-Protein Interactions (PPI) are the binding relationships between proteins.
- GRNsight is an open source tool that allows for easier visualization of GRNs and PPIs.
- We transitioned our database source from YeastMine to AllianceMine.
- GRNsight has improved PPI support for node coloring.
- GRNs have directed edges to indicate regulatory relationships (activation or repression).
- PPIs have undirected edges to indicate physical binding.
- New graph functionality allows users to restrict the graph to the viewport.
- GRNsight switched CI tools from Travis CI to Github Actions.

Edge weight values are indicated by color, thickness, and end marker



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GRNsight v7.2.0

Welcome to GRNsight. Please select a file to upload.

Network Layout Node Edge View Export Help Demo Demo #2: Weighted GRN (15 genes, 28 edges, Dahlquist Lab unpublished data)

15 nodes
28 edges

Network

Layout

Node

Edge

☒ Enable Edge Coloring

Hide/Show Edge Weights

☐ Show With Mouse Over

☐ Always Show Edge Weights

☒ Never Show Edge Weights

Edge Weight Normalization

Factor (0.0001-1000):

5.21

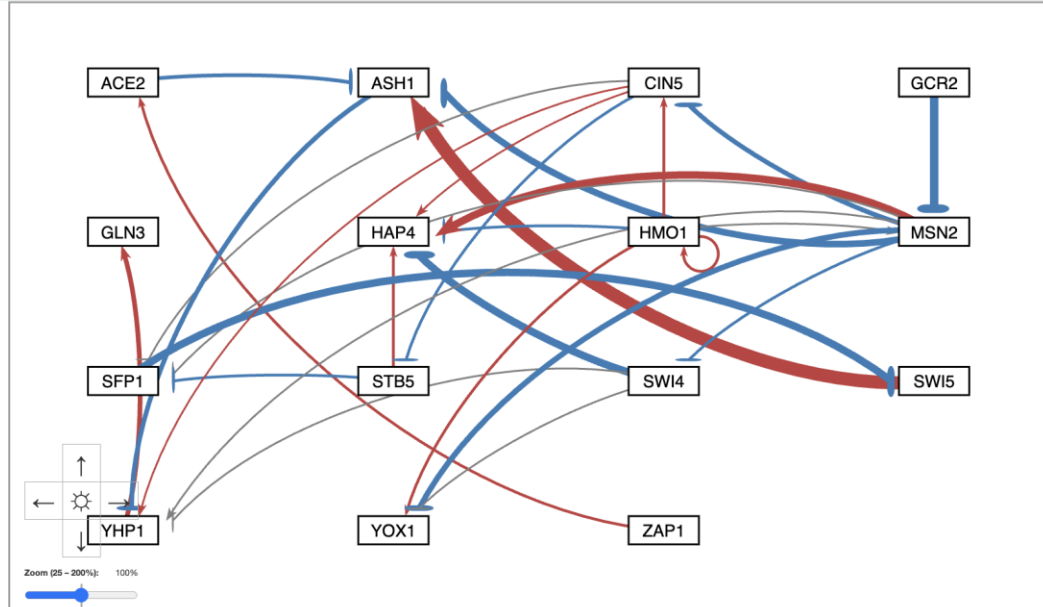
Set Factor

Reset Factor

Gray Threshold (0-100%): 5%

☐ Show Gray Edges as Dashed

View



Numerical edge weight values can be displayed on the graph



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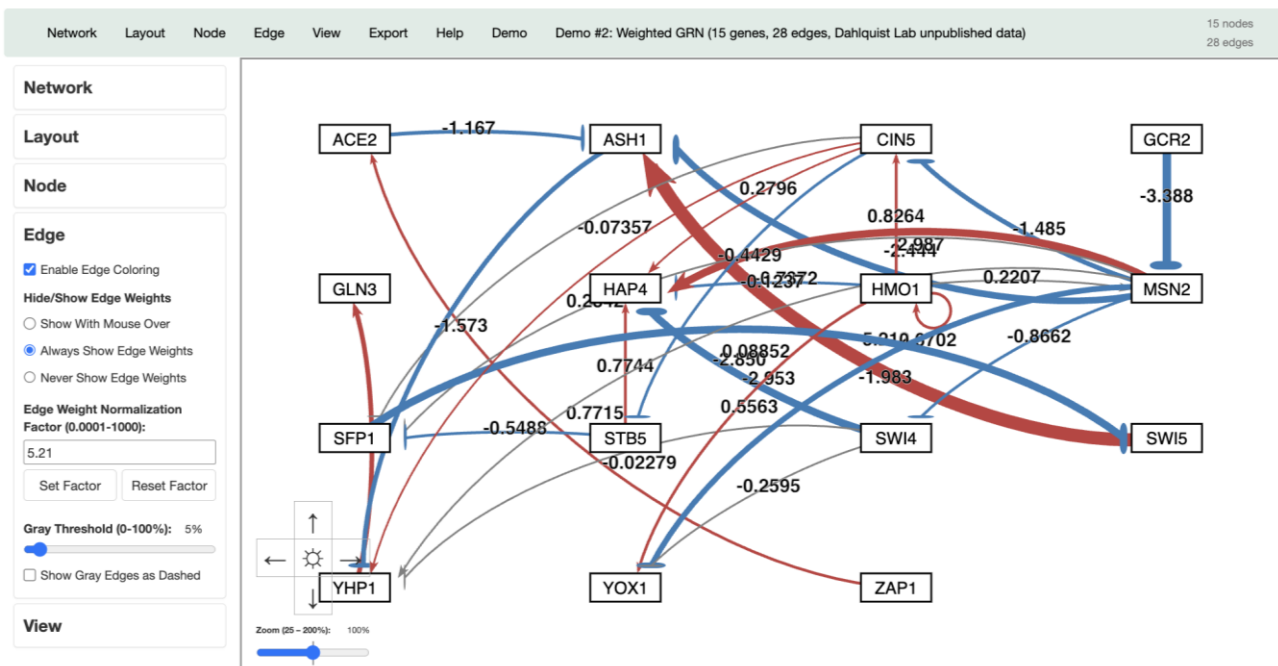
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GRNsight v7.2.0

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Edge thicknesses are normalized to the largest magnitude weight value



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GRNsight v7.2.0

Normalization Factor = 5.21

Welcome to GRNsight. Please select a file to upload.

Network Layout Node Edge View Export Help Demo Demo #2: Weighted GRN (15 genes, 28 edges, Dahlquist Lab unpublished data)

15 nodes
28 edges

Network

Layout

Node

Edge

☒ Enable Edge Coloring

Hide/Show Edge Weights

☐ Show With Mouse Over

☐ Always Show Edge Weights

☒ Never Show Edge Weights

Edge Weight Normalization
Factor (0.0001-1000):

5.21

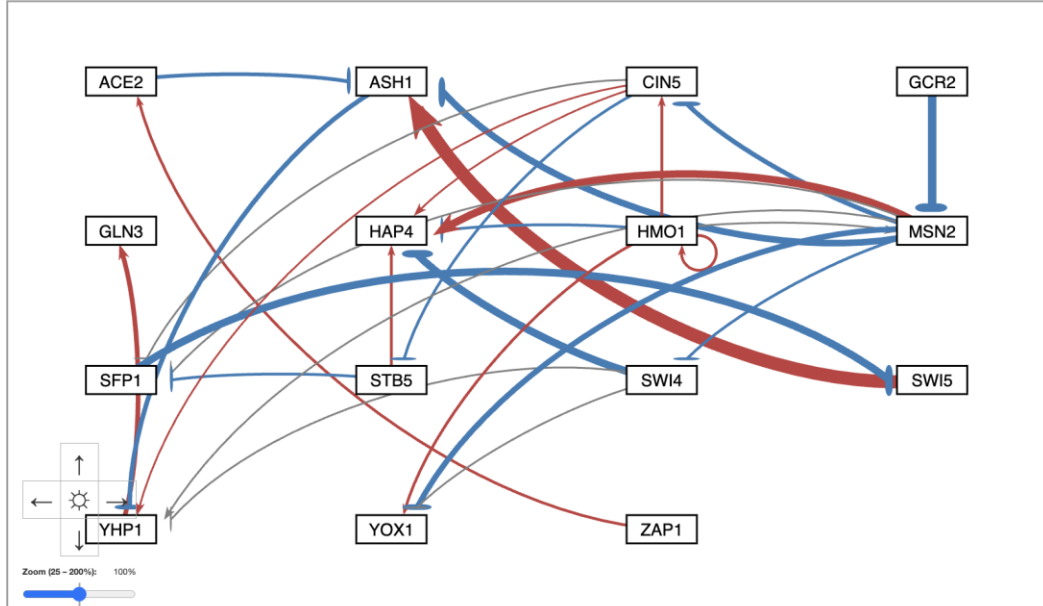
Set Factor

Reset Factor

Gray Threshold (0-100%): 5%

☐ Show Gray Edges as Dashed

View



Normalization factors can be modified to allow fair visual comparison between different graphs



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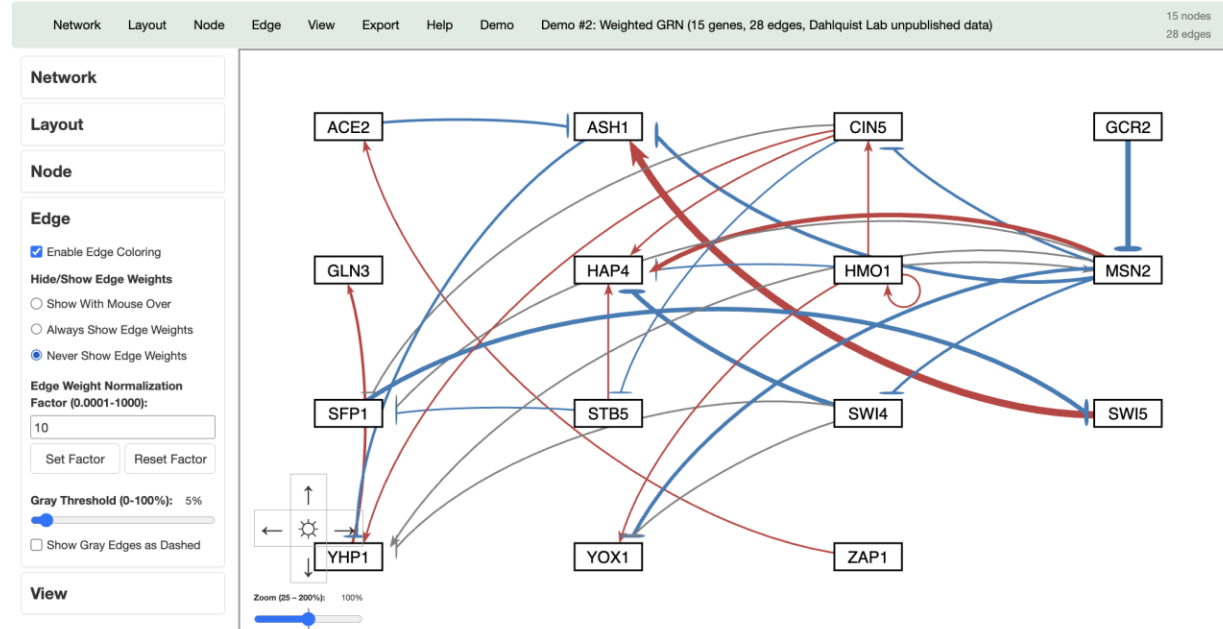
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GRNsight v7.2.0

Normalization Factor = 10

Welcome to GRNsight. Please select a file to upload.



Normalization factors can be modified to allow fair visual comparison between different graphs



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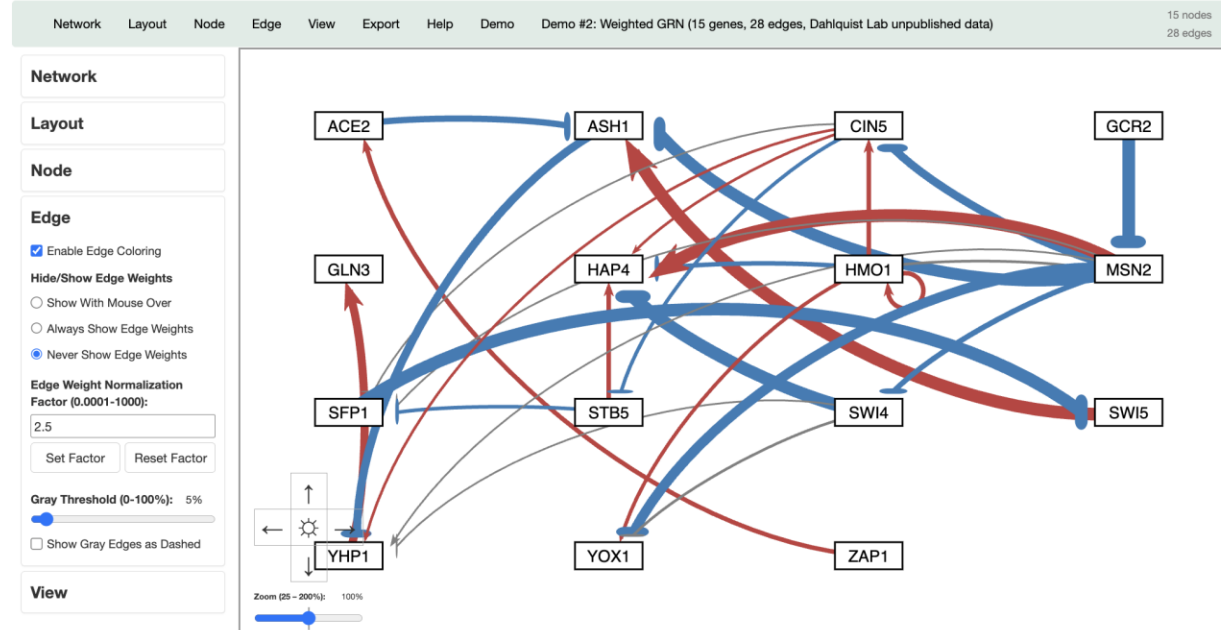
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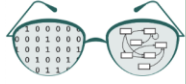
GRNsight v7.2.0

Normalization Factor = 2.5

Welcome to GRNsight. Please select a file to upload.



Edges with normalized weight values within 5% of zero are displayed as gray to minimize their visual strength



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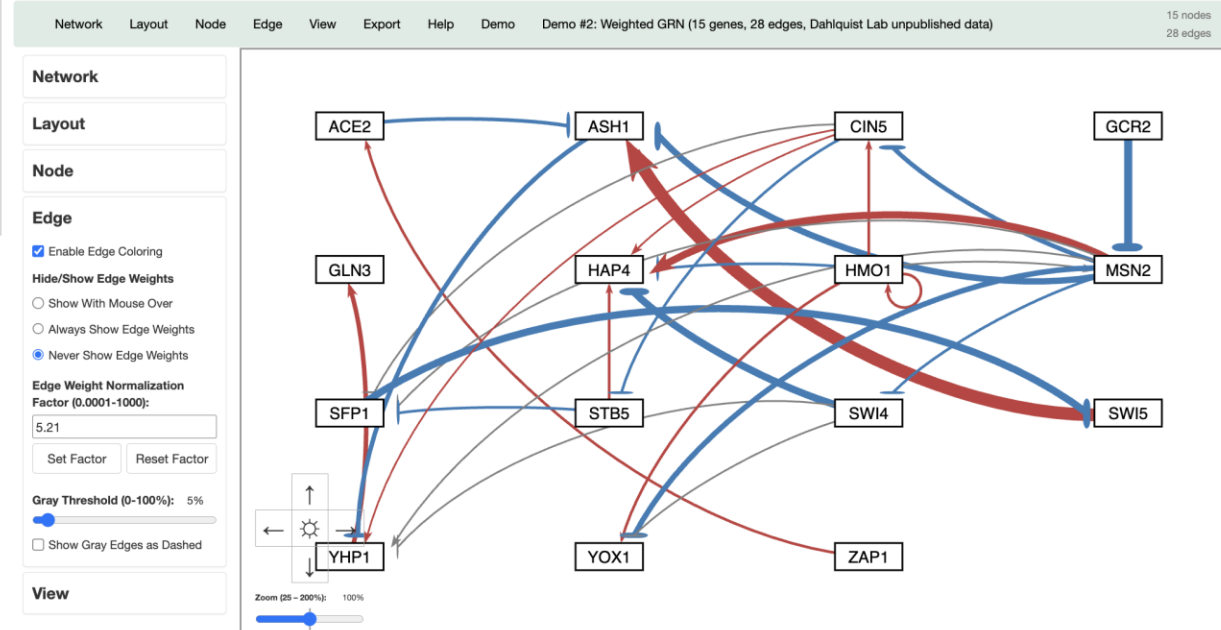
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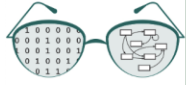
GRNsight v7.2.0

Welcome to GRNsight. Please select a file to upload.

Normalization Factor = 5.21
Gray Threshold = 5%



Users can change the gray threshold with a slider



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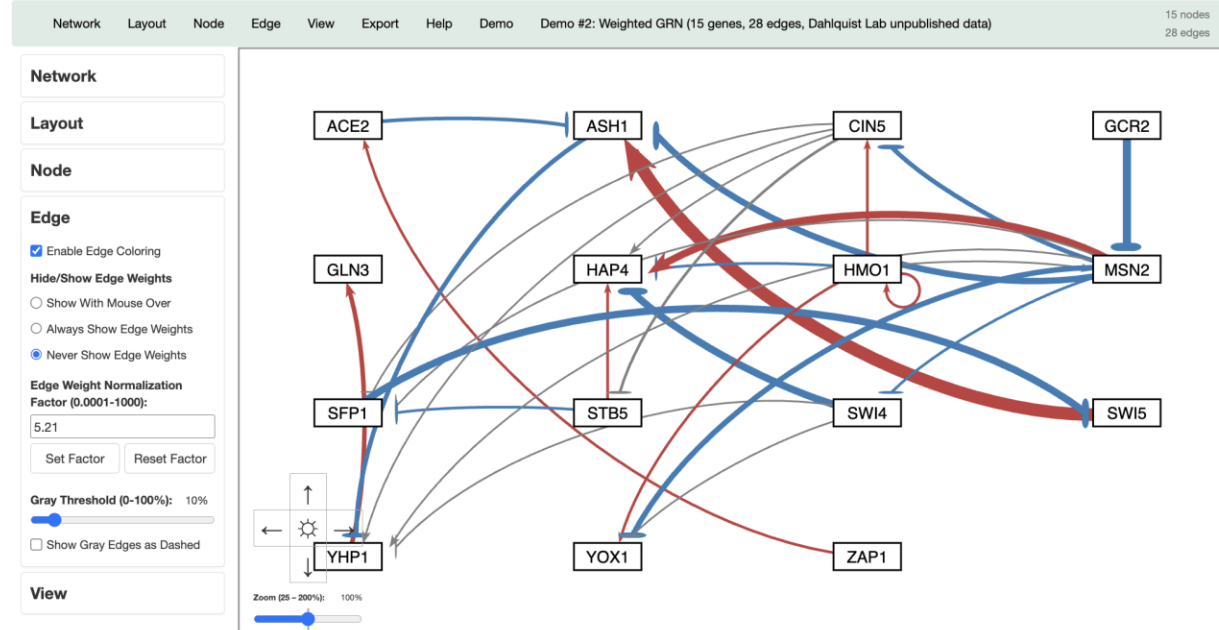
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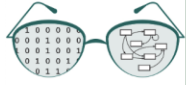
GRNsight v7.2.0

Welcome to GRNsight. Please select a file to upload.

Normalization Factor = 5.21
Gray Threshold = 10%



Users can change the gray threshold with a slider



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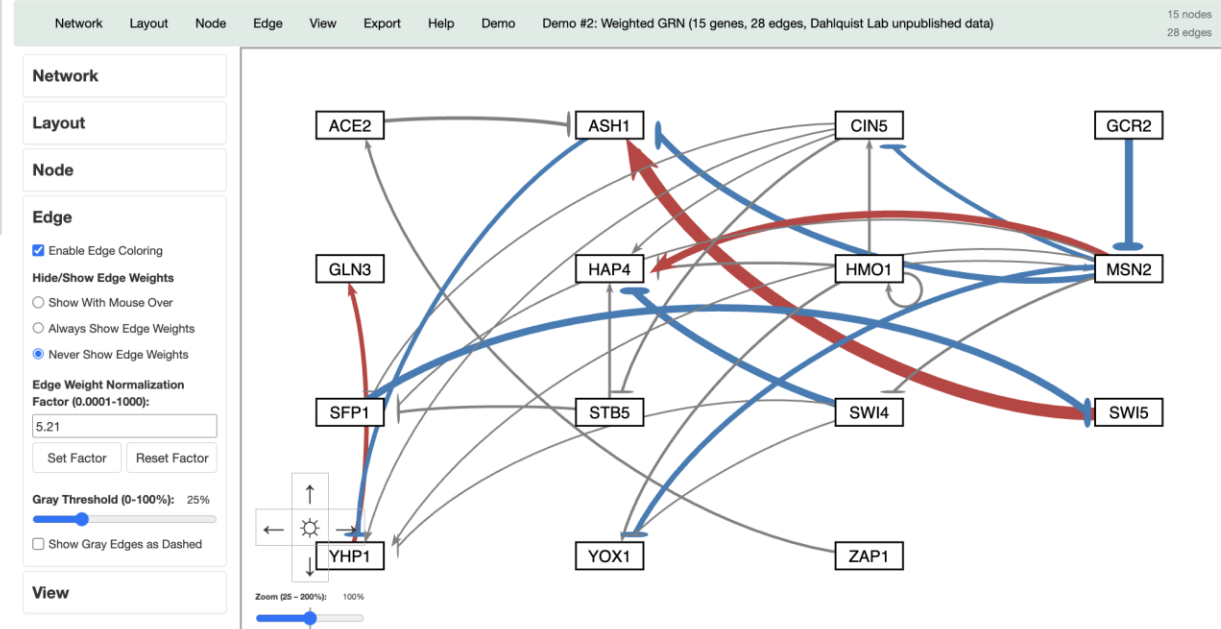
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GRNsight v7.2.0

Welcome to GRNsight. Please select a file to upload.

Normalization Factor = 5.21
Gray Threshold = 25%



Users can change the gray threshold with a slider



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GRNsight v7.2.0

Welcome to GRNsight. Please select a file to upload.

Normalization Factor = 5.21
Gray Threshold = 50%

Network Layout Node Edge View Export Help Demo Demo #2: Weighted GRN (15 genes, 28 edges, Dahlquist Lab unpublished data)

15 nodes
28 edges

Network

Layout

Node

Edge

☒ Enable Edge Coloring

Hide/Show Edge Weights

☐ Show With Mouse Over

☐ Always Show Edge Weights

☒ Never Show Edge Weights

Edge Weight Normalization
Factor (0.0001-1000):

5.21

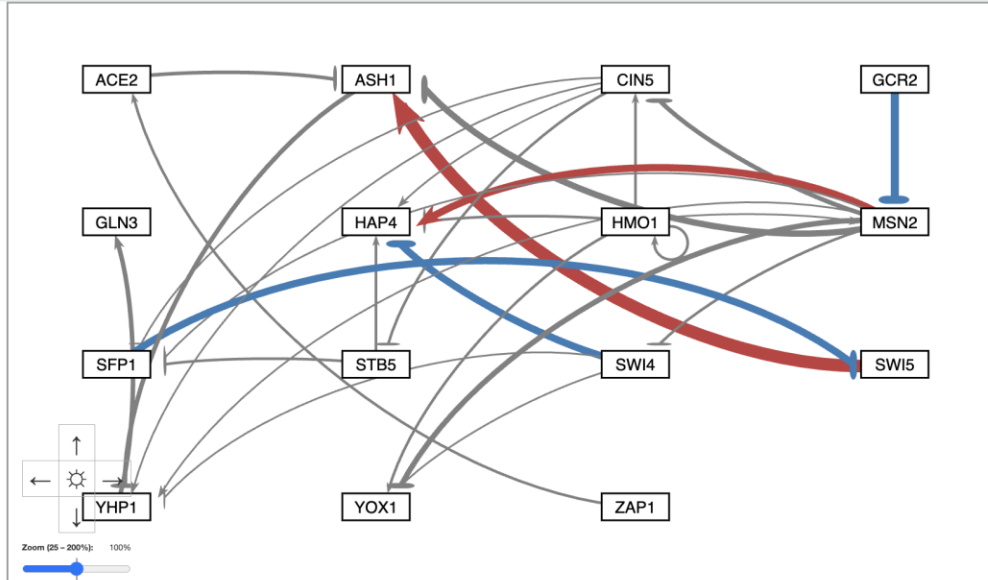
Set Factor

Reset Factor

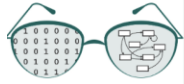
Gray Threshold (0-100%): 50%

☐ Show Gray Edges as Dashed

View



Gray edges can also be shown as dashed lines to facilitate visualization by color blind users



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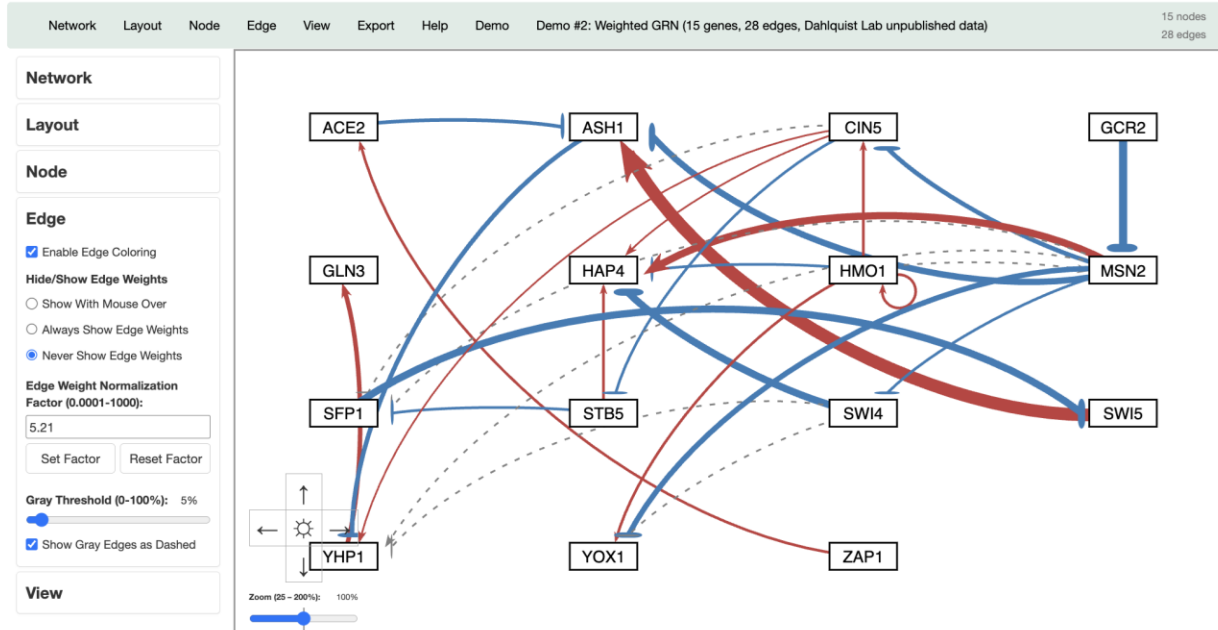
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GRNsight v7.2.0

Welcome to GRNsight. Please select a file to upload.

Normalization Factor = 5.21
Gray Threshold = 5%



GRNsight displays PPI networks as undirected graphs for they are just binding relationships, not regulatory



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12/4/2004

GRNsight v7.2.0

Welcome to GRNsight. Please select a file to upload.

Network Layout Node Edge View Export Help Demo Demo #5: PPI (18 proteins, 81 edges)

18 nodes
81 edges

Network

Network Source

Demo

Demo #5: PPI (18 pro

Open File

Load from Database

Reload

Network Mode

Protein-Protein Physica

Species:

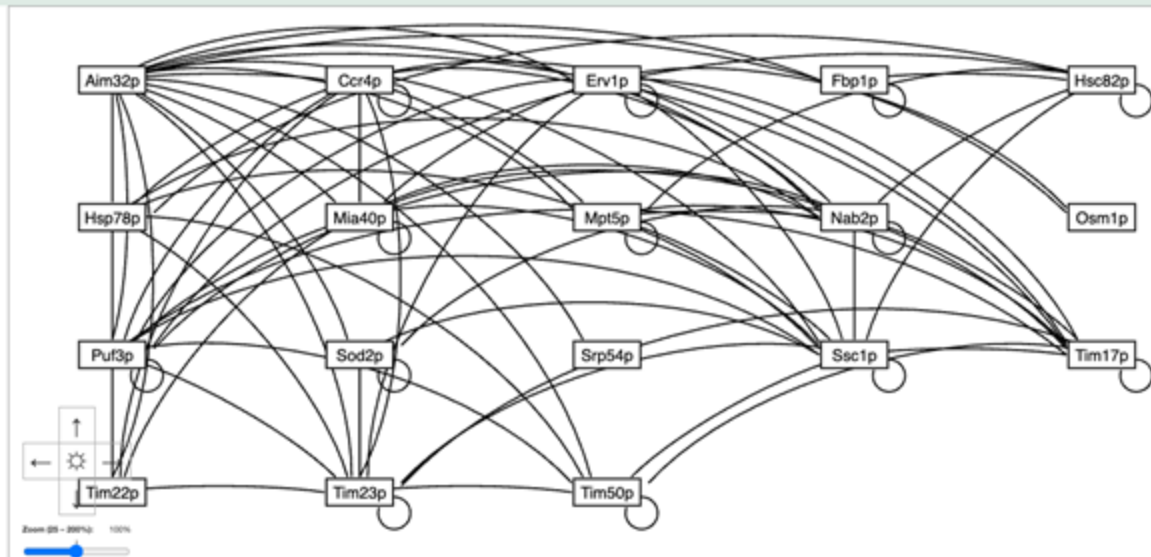
Saccharomyces cerevisiae

Layout

Node

Edge

View



Outline

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- GRNs have directed edges to indicate regulatory relationships (activation or repression).
- PPIs have undirected edges to indicate physical binding.
- **New graph functionality allows users to restrict the graph to the viewport.**
- GRNsight switched CI tools from Travis CI to Github Actions.

Users can choose between a small, medium, or large viewport size



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Last modified:
11/18/2025

Beta v7.3.4

Disclaimer: This is the beta version of GRNsight for live testing of new functionality. For the most stable version, please go to the GRNsight home page.

Network Layout Node Edge View Export Help Demo Demo #3: Unweighted GRN (21 genes, 31 edges) 21 nodes 31 edges

Network

Layout

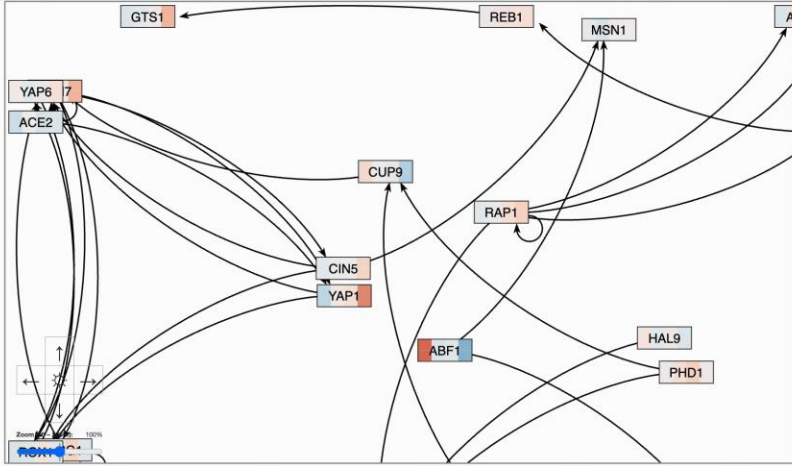
Node

Edge

View

Size

- ☒ Small (1104 X 648 pixels)
- ☐ Medium (1414 X 840 pixels)
- ☐ Large (1920 X 1080 pixels)
- ☐ Fit to window
- ☐ Restrict graph to viewport



Zoom: 100%

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Users can drag, zoom, and move the graph with nodes staying within application's bounding box



GRNsight

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Network

Layout

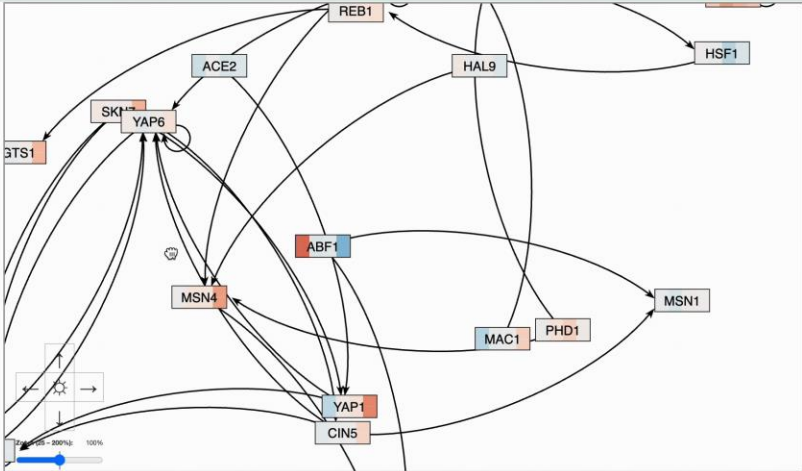
Node

Edge

View

Size

- ☒ Small (1104 X 648 pixels)
- ☐ Medium (1414 X 840 pixels)
- ☐ Large (1820 X 1080 pixels)
- ☐ Fit to window
- ☐ Restrict graph to viewport



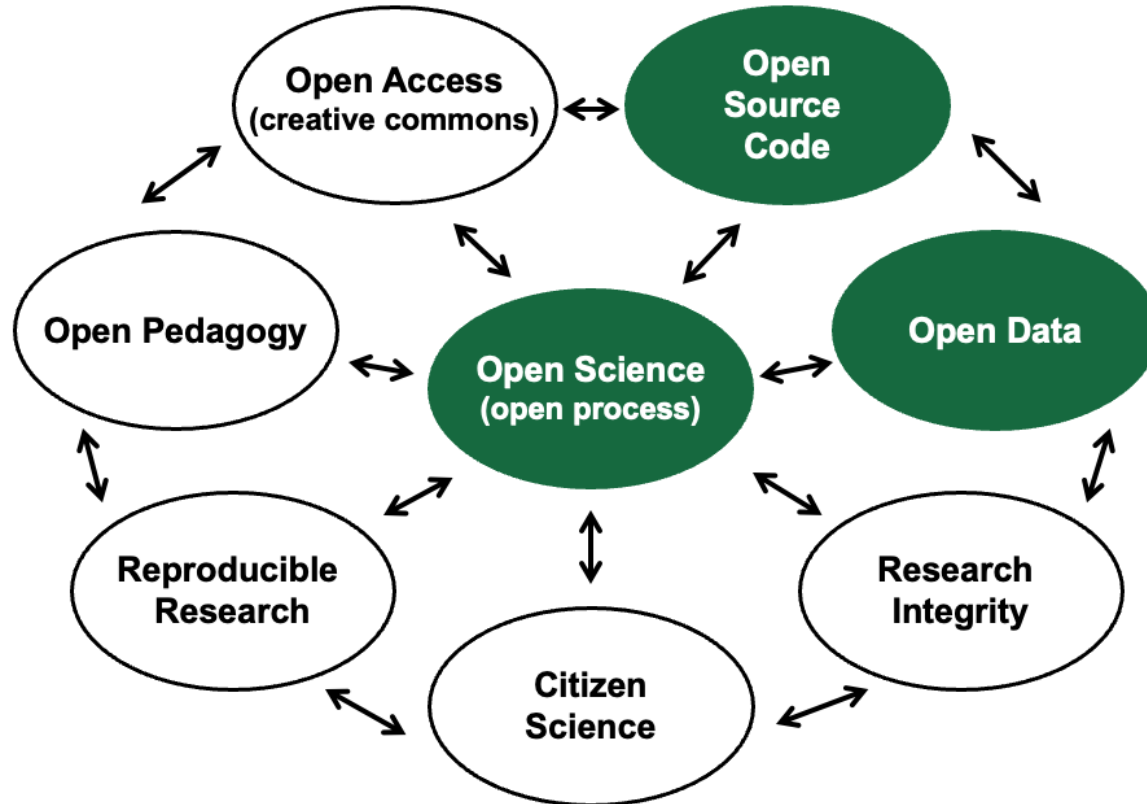
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- GRNsight switched CI tools from Travis CI to Github Actions.

GRNsight is part of an open science ecosystem



With thanks to John Jungck

All GRNsight development occurs in an open source repository

The screenshot shows the GitHub interface for the **dondi / GRNsight** repository. The repository is public and has 6 watchers, 8 forks, and 17 stars. The main branch is **main**, with 15 branches and 28 tags. The repository contains a file tree with folders like **.github/workflows**, **assets/images**, **database**, **server**, **test-files**, **test**, and **web-client**, and files like **.eslintrc**, **.gitignore**, **.npmignore**, **.travis.yml**, **CITATION.md**, **CODE_OF_CONDUCT.md**, **LICENSE.txt**, and **Makefile**. The right sidebar shows the **About** section with a description, links to the README, license, and code of conduct, and the **Releases** section showing the latest release **v7.2.0** from December 4, 2024. The **Packages** section shows no published packages, and the **Contributors** section shows 28 contributors.

dondi / GRNsight Public

Watch 6 Fork 8 Star 17

main 15 Branches 28 Tags

Go to file Add file Code

dondi Version bump. ✓ c19c00b · last year 3,508 Commits

.github/workflows	Merge pull request #1155 from dondi/#1145-gain-compati...	last year
assets/images	added new file structure for Documents	9 years ago
database	Remove testing schema that I forgot to delete	last year
server	#1030 Remove lint changes and only change the variable ...	last year
test-files	Add demo 5 Protein-protein-physical-interaction	2 years ago
test	#1197 Modify alpha value on optimization_parameters fro...	last year
web-client	#1144 Remove unused import.	last year
.eslintrc	Merged changes made by Eileen and Jen	7 years ago
.eslintrc.yml	eslint fix	2 years ago
.gitignore	#1020: completed protein database schema, generation s...	2 years ago
.npmignore	Fixed errors in error and warning lists	10 years ago
.travis.yml	trying a before install to choose which npm version we wa...	4 years ago
CITATION.md	fix link format	8 years ago
CODE_OF_CONDUCT.md	add code of conduct	9 years ago
LICENSE.txt	Assign 3-clause BSD license to GRNsight.	11 years ago
Makefile	Add starter Makefile for mocha.	11 years ago

About

Web app and service for modeling and visualizing gene regulatory networks.

dondi.github.io/GRNsight

- Readme
- BSD-3-Clause license
- Code of conduct
- Cite this repository
- Activity
- 17 stars
- 6 watching
- 8 forks

Report repository

Releases 23

v7.2.0 (Latest) on Dec 4, 2024

+ 22 releases

Packages

No packages published

[Publish your first package](#)

Contributors 28

All source code can be viewed in the Code tab

The screenshot shows the GitHub interface for the repository **dondi / GRNsight**. The **Code** tab is selected, displaying a file tree on the left and a commit history table. The right sidebar contains repository metadata, including a description, links to the README and license, and release information.

Repository Header: dondi / GRNsight (Public)

Navigation: Code (selected), Issues (50), Pull requests (2), Actions, Projects (2), Wiki, Security (110), Insights, Settings.

Repository Stats: Watch (6), Fork (8), Star (17).

File Tree:

- .github/workflows
- assets/images
- database
- server
- test-files
- test
- web-client
- .eslintrc
- .eslintrc.yml
- .gitignore
- .npmignore
- .travis.yml
- CITATION.md
- CODE_OF_CONDUCT.md
- LICENSE.txt
- Makefile

Commit History Table:

Commit	Message	Time
dondi	Version bump. ✓	c19c00b · last year · 3,508 Commits
	Merge pull request #1155 from dondi/#1145-gain-compati...	last year
	added new file structure for Documents	9 years ago
	Remove testing schema that I forgot to delete	last year
	#1030 Remove lint changes and only change the variable ...	last year
	Add demo 5 Protein-protein-physical-interaction	2 years ago
	#1197 Modify alpha value on optimization_parameters fro...	last year
	#1144 Remove unused import.	last year
	Merged changes made by Eileen and Jen	7 years ago
	eslint fix	2 years ago
	#1020: completed protein database schema, generation s...	2 years ago
	Fixed errors in error and warning lists	10 years ago
	trying a before install to choose which npm version we wa...	4 years ago
	fix link format	8 years ago
	add code of conduct	9 years ago
	Assign 3-clause BSD license to GRNsight.	11 years ago
	Add starter Makefile for mocha.	11 years ago

About: Web app and service for modeling and visualizing gene regulatory networks.
Link: dondi.github.io/GRNsight

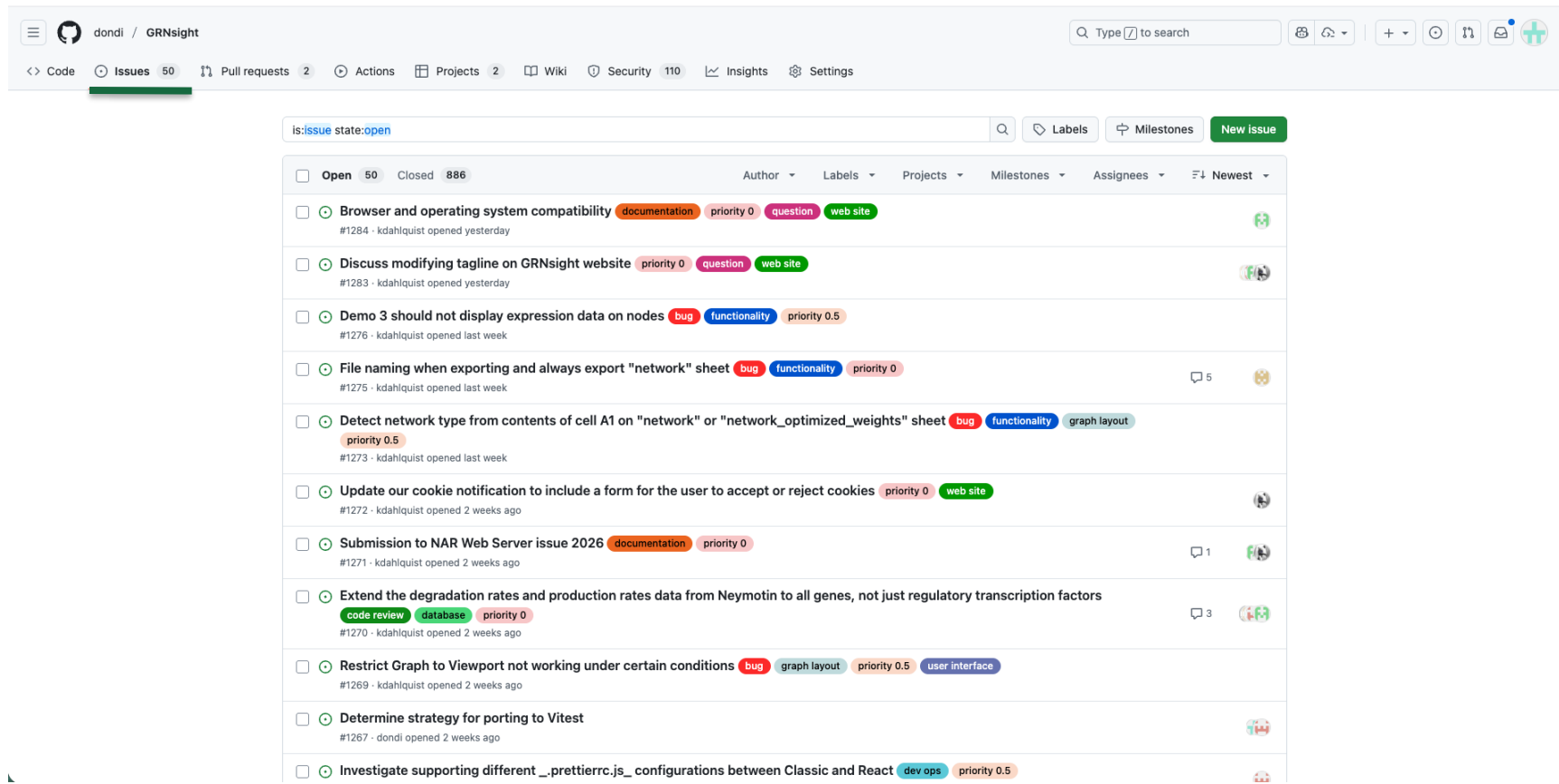
Repository Info: README, BSD-3-Clause license, Code of conduct, Cite this repository, Activity, 17 stars, 6 watching, 8 forks.

Releases: 23 releases. Latest: v7.2.0 (on Dec 4, 2024). + 22 releases.

Packages: No packages published. [Publish your first package](#)

Contributors: 28 contributors.

All feature requests and bugs are reported in the Issues tab



The screenshot shows the GitHub interface for the **dondi / GRNsight** repository. The **Issues** tab is active, displaying a list of 50 open issues. The issues are filtered by `is:issue state:open`. The list includes various feature requests and bugs, each with a title, issue number, author, and labels.

Issue Title	Issue Number	Author	Labels
Browser and operating system compatibility	#1284	kdahlquist	documentation, priority 0, question, web site
Discuss modifying tagline on GRNsight website	#1283	kdahlquist	priority 0, question, web site
Demo 3 should not display expression data on nodes	#1276	kdahlquist	bug, functionality, priority 0.5
File naming when exporting and always export "network" sheet	#1275	kdahlquist	bug, functionality, priority 0
Detect network type from contents of cell A1 on "network" or "network_optimized_weights" sheet	#1273	kdahlquist	priority 0.5, bug, functionality, graph layout
Update our cookie notification to include a form for the user to accept or reject cookies	#1272	kdahlquist	priority 0, web site
Submission to NAR Web Server issue 2026	#1271	kdahlquist	documentation, priority 0
Extend the degradation rates and production rates data from Neymotin to all genes, not just regulatory transcription factors	#1270	kdahlquist	code review, database, priority 0
Restrict Graph to Viewport not working under certain conditions	#1269	kdahlquist	bug, graph layout, priority 0.5, user interface
Determine strategy for porting to Vitest	#1267	dondi	
Investigate supporting different <code>_prettierrc.js_</code> configurations between Classic and React			dev ops, priority 0.5

Developers create pull requests in the Pull Request tab when their code is ready to review and merge

dondi / GRNsight

Q Type [Z] to search

<> Code

Issues 50

Pull requests 2

Actions

Projects 2

Wiki

Security 110

Insights

Settings

Label issues and pull requests for new contributors

Dismiss

Now, GitHub will help potential first-time contributors [discover issues](#) labeled with [good first issue](#)

Go to Labels

Filters

Q is:pr is:open

Labels 34

Milestones 0

New pull request

☐

2 Open

✓ 347 Closed

Author

Label

Projects

Milestones

Reviews

Assignee

Sort

☐

1265: Configure Prod React App to Access Server

✓

#1279 opened last week by ceciliazaragoza · Draft

☐

Added wrap-around styling for demo name in export to excel window

✓

#1263 opened 3 weeks ago by MilkaZek

4

💡 ProTip!

Filter pull requests by the default branch with [base:main](#).

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This is where Continuous Integration (CI) plays an important role

 **Added wrap-around styling for demo name in export to excel window #1263**
MilkaZek wants to merge 3 commits into [beta](#) from [milka-export-bra...](#)

 **coveralls commented yesterday**

coverage **80%**

coverage: 80.06%. remained the same
when pulling [ef3f34b](#) on milka-export-branch
into [569311d](#) on beta.

 **This branch has not been deployed**
No deployments



 **All checks have passed**
5 successful checks

  [coverage/coveralls](#) — Coverage remained the same at 80.06%

  [coverage/coveralls \(push\)](#) — Coverage remained the same at 80.06%

  [Node.js CI / build \(18.x\) \(push\)](#) Successful in 1m

  [Node.js CI / build \(20.x\) \(push\)](#) Successful in 1m

  [Node.js CI / build \(22.x\) \(push\)](#) Successful in 1m

 **No conflicts with base branch**
Merging can be performed automatically.

Merge pull request

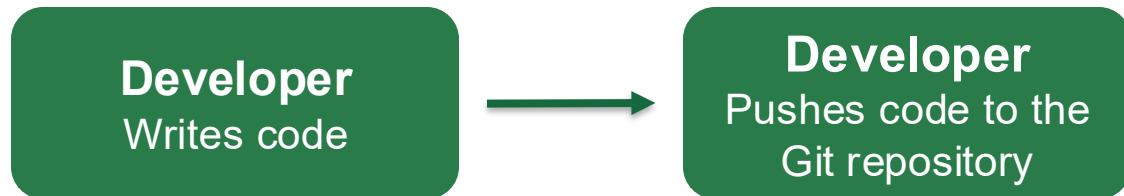
You can also merge this with the command line. [View command line instructions.](#)

Still in progress? [Convert to draft](#)

Continuous Integration is the practice of developers frequently merging code into a central repository

Developer
Writes code

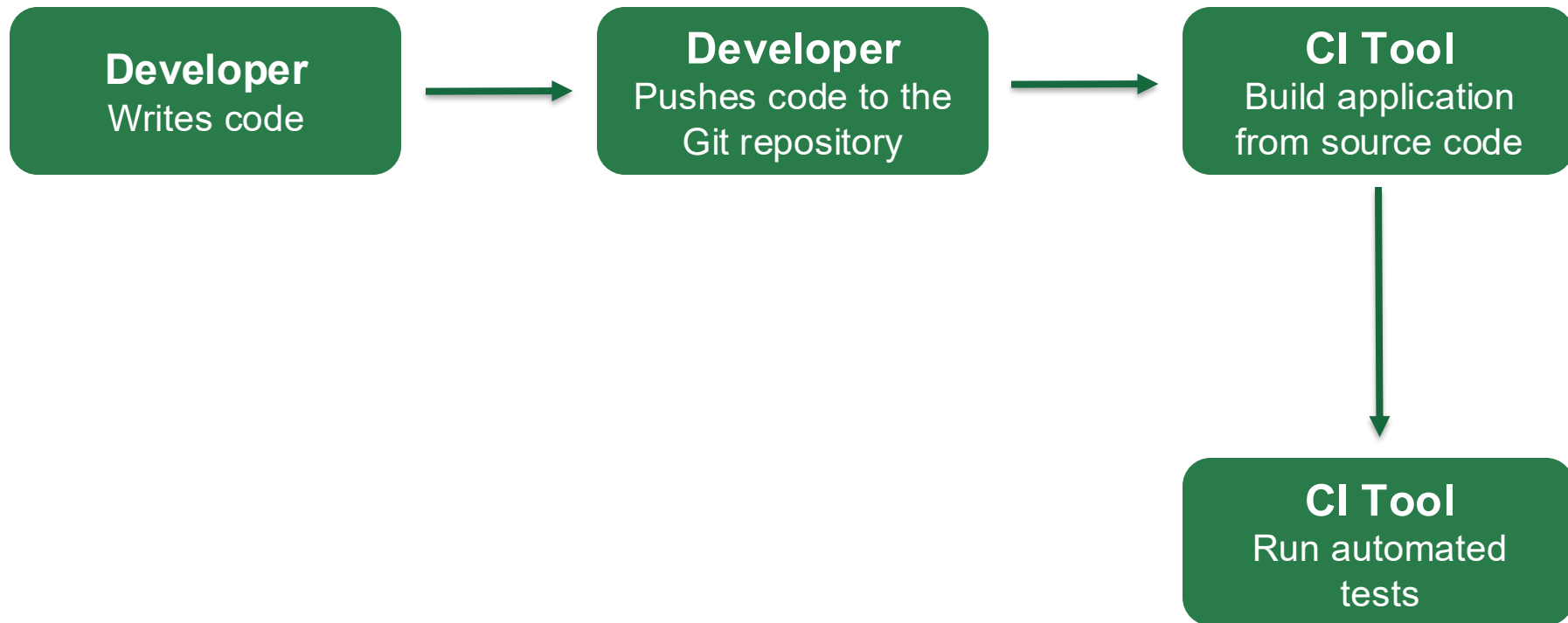
Continuous Integration is the practice of developers frequently merging code into a central repository



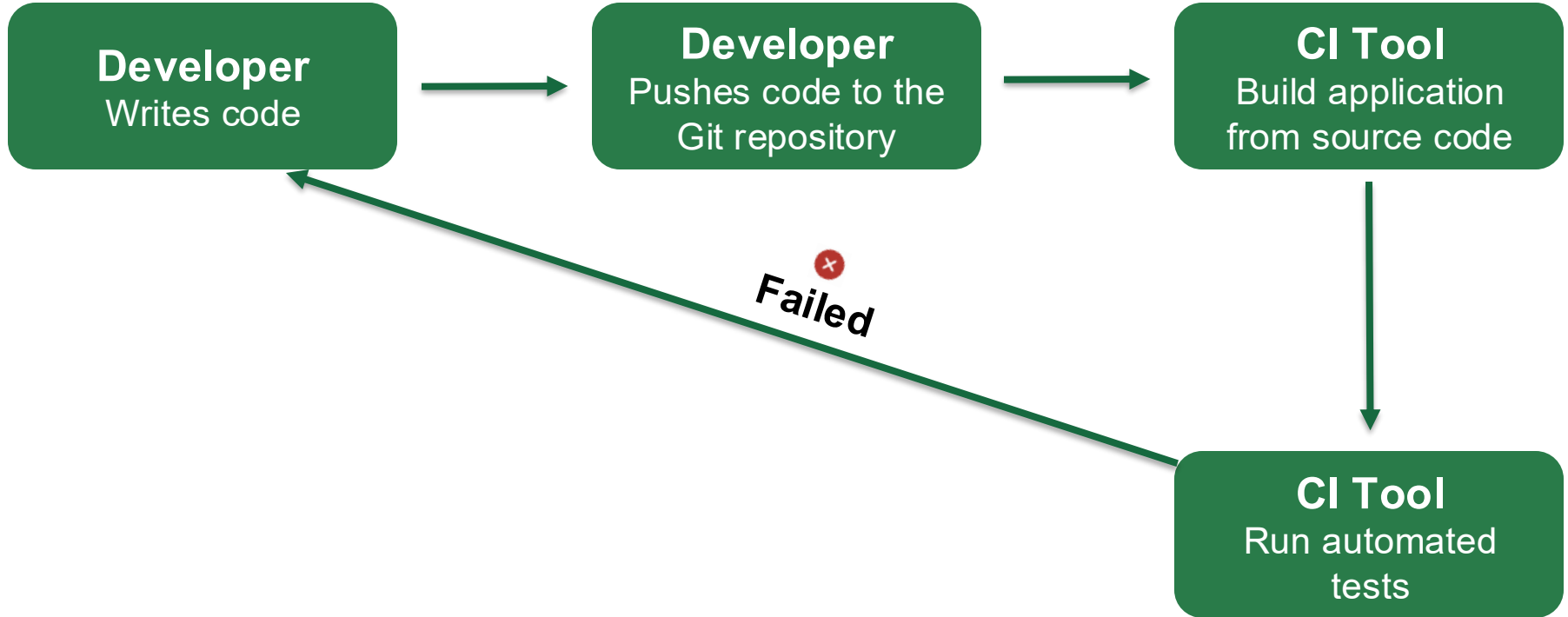
Every code push triggers a build



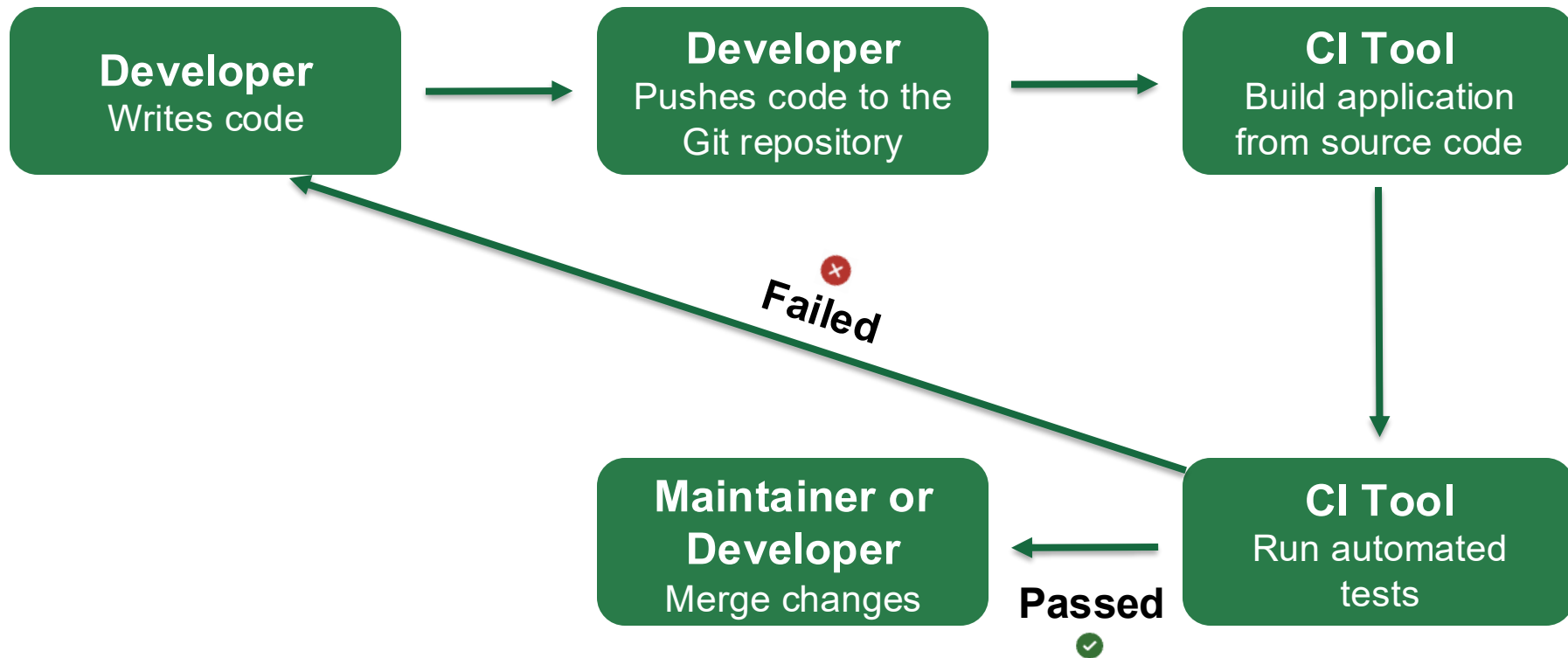
The build is validated by running automated tests



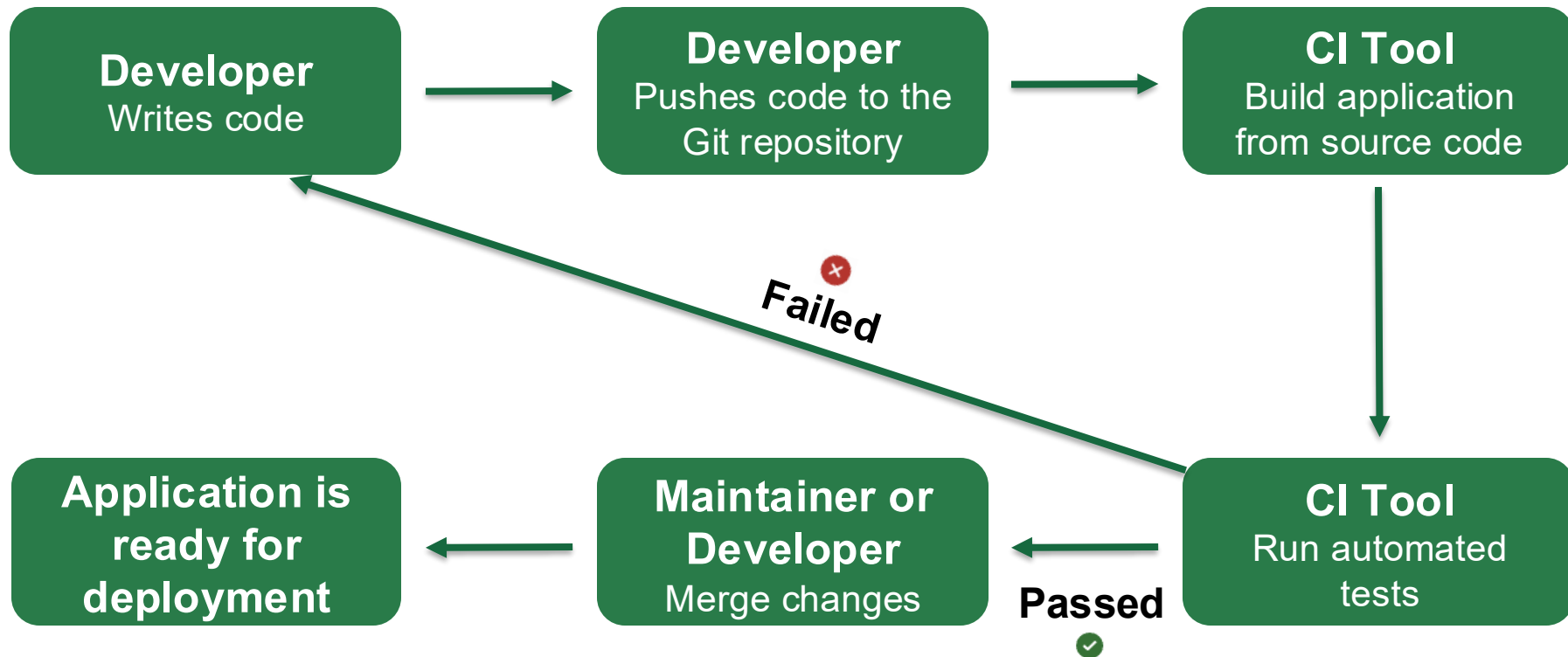
If a problem is detected, developers need to make changes and edit their code before making another commit



If all automated tests passed, the maintainer or developer merges the changes



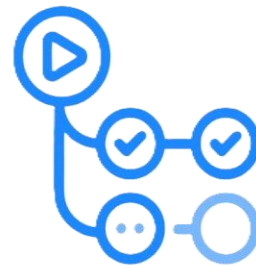
After all changes get merged, the application is ready for deployment



V7 - GRNsight migrated from Travis CI to Github Actions for CI tools due to Travis' user limit for their free plan



Travis CI



GitHub Actions

Github Actions is native to Github so no external connection is needed

The screenshot shows the GitHub Actions interface for the repository 'dondi / GRNsight'. The top navigation bar includes links to Code, Issues (50), Pull requests (2), Actions (selected), Projects (2), Wiki, Security (110), Insights, and Settings. A search bar is located on the right. The left sidebar shows the 'Actions' section with a 'New workflow' button and a list of workflow categories: Copilot code review, Node.js CI, pages-build-deployment, Management, Caches, Deployments, Attestations, Runners, Usage metrics, and Performance metrics. The main content area is titled 'All workflows' and shows 'Showing runs from all workflows'. A search bar for 'Filter workflow runs' is present. Below this, a table lists 345 workflow runs. The table has columns for Event, Status, Branch, and Actor. The runs are listed in descending order of time.

Event	Status	Branch	Actor
pages build and deployment	Success	gh-pages	dondi
Version bump.	Success	beta	dondi
Merge pull request #1285 from dondi/amelie-1174	Success	beta	dondi
Merge pull request #1282 from dondi/maika-1223	Success	beta	dondi
Merge pull request #1280 from dondi/maika-1270	Success	beta	dondi
Fix linting errors with prettier	Success	amelie-1174	Amelie1253
Fix linting errors with prettier	Failure	amelie-1174	Amelie1253
cleaned up code	Failure	amelie-1174	Amelie1253
wrapped up functionality for export issue 1174	Failure	amelie-1174	Amelie1253

Github Actions also allows for testing against multiple versions of the Node.js interpreter

The screenshot shows a GitHub repository named 'dondli / GRNsight'. The 'Actions' tab is selected, displaying a workflow run titled 'separated some funcs into different files #251'. The workflow is for Node.js 18.x and has succeeded 4 days ago in 1m 22s. The left sidebar shows a list of jobs: 'build (18.x)', 'build (20.x)', and 'build (22.x)'. The main panel shows the 'build (18.x)' job details, including a search bar for logs and a list of steps with their durations.

Step	Duration
Set up job	2s
Checkout repository	1s
Install dependencies for canvas	37s
Set up Node.js 18.x	5s
Install dependencies	13s
Run Linter	4s
Run Build	8s
Run Tests with Coverage	15s
Check Coverage File	8s
Upload Coverage to Coveralls	1s
Post Set up Node.js 18.x	1s
Post Checkout repository	9s
Complete job	0s

Green check indicates the code passed all the tests

The screenshot shows a GitHub Actions workflow run for the repository 'dondli / GRNsight'. The workflow is named 'separated some funcs into different files #251' and is currently in a 'Completed' state, indicated by a green checkmark. The workflow is triggered by a 'Node.js CI' event.

The workflow run is titled 'build (18.x)' and succeeded 4 days ago in 1m 22s. The workflow consists of several jobs, each with a green checkmark indicating success:

- build (18.x) - 2s
- build (20.x) - 1s
- build (22.x) - 37s

The 'build (18.x)' job is expanded, showing a list of steps, each with a green checkmark indicating success:

- Set up job - 2s
- Checkout repository - 1s
- Install dependencies for canvas - 37s
- Set up Node.js 18.x - 5s
- Install dependencies - 13s
- Run Linter - 4s
- Run Build - 8s
- Run Tests with Coverage - 15s
- Check Coverage File - 8s
- Upload Coverage to Coveralls - 1s
- Post Set up Node.js 18.x - 1s
- Post Checkout repository - 9s
- Complete job - 9s

The left sidebar shows the workflow run details, including a summary, jobs, and run details. The 'Jobs' section is expanded, showing the 'build (18.x)' job as the first step. The 'Run details' section is also expanded, showing the 'Usage' and 'Workflow file' sections.

A red 'X' appears if the test failed

Summary

Jobs

build (18.x)

build (20.x)

build (22.x)

Run details

Usage

Workflow file

build (22.x)

failed 5 hours ago in 43s

Search logs

Explain error

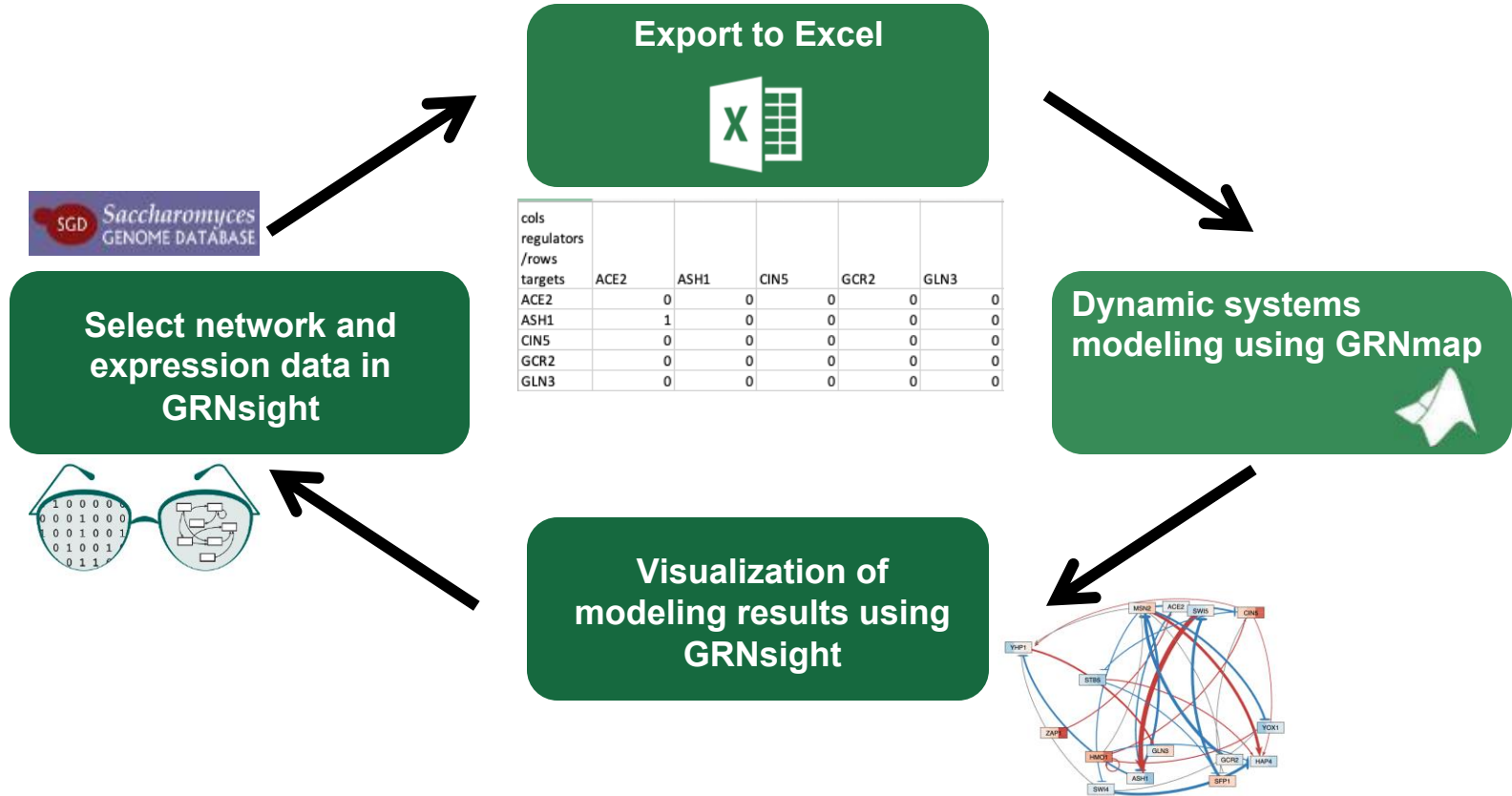


>	✓	Install dependencies for canvas	22s
>	✓	Set up Node.js 22.x	0s
>	✓	Install dependencies	14s
▼	✗	Run Linter	3s
1		▶ Run npm run lint	
4			
5		> grnsight@7.2.6 lint	
6		> eslint "**/*.js"	
7			
8			
9		/home/runner/work/GRNsight/GRNsight/web-client-classic/public/js/upload.js	
10	Error:	525:50 error Insert `.`	prettier/prettier
11	Error:	527:17 error Replace `_.toLowerCase().startsWith(source` with `_.toLowerCase().startsWith(source`	prettier/prettier
12	Error:	529:1 error Replace `.....?`checked="true".....` with `.....?`checked="true".....`	prettier/prettier
13	Error:	537:19 error Insert `.`	prettier/prettier
14	Error:	736:45 error Insert `.....`	prettier/prettier
15	Error:	752:14 error Delete `.....`	prettier/prettier
16	Error:	774:87 error Delete `.....`	prettier/prettier
17			
18		✖ 7 problems (7 errors, 0 warnings)	
19		7 errors and 0 warnings potentially fixable with the `--fix` option.	
20			
21	Error:	Process completed with exit code 1.	
✓		Run Build	0s
✓		Run Tests with Coverage	0s
✓		Check Coverage File	0s
✓		Upload Coverage to Coveralls	0s
✓		Post Set up Node.js 22.x	0s
>	✓	Post Checkout repository	0s
>	✓	Complete job	0s

Summary of improvements to GRNsight v7

- Allows for easier visualization of GRNs and PPIs
- Supports node coloring and different edge display factors
- Provides viewport controls
- Streamlines development workflow

GRNsight v7 is an integral part of a data analysis workflow for biologists who model GRNs and PPIs



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<https://dondi.github.io/GRNsight/>



<https://github.com/dondi/GRNsight>

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